

How Stock Market Participants Use Generative Artificial Intelligence: Evidence from User-Platform Interaction Data

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ABSTRACT

This paper provides descriptive evidence on how stock market participants use Generative Artificial Intelligence (GenAI) to process investment-related information. Using a data set of 1.7 million stock-related queries from one of China's largest GenAI platforms during the first half of 2024, we document that user queries address a wide range of topics and tasks and vary systematically with usage intensity and financial sophistication. Query activity increases around corporate disclosure events, but these increases largely track contemporaneous media coverage. We also find evidence consistent with a substi-

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tution between the informativeness of voluntary managerial disclosures and investors' reliance on GenAI. Continued platform engagement more likely follows answers that are concise and contain directionally accurate trading signals. Over time, users' subsequent queries increasingly reflect the specificity and financial terminology present in earlier GenAI answers. At the market level, GenAI usage is associated with higher measures of informed trading and lower liquidity, while aggregated sentiment in GenAI-generated answers correlates with same-day abnormal returns, particularly when user feedback is positive. Overall, our findings offer insights into early-stage GenAI adoption by retail investors and inform discussions on how GenAI shapes information processing in financial markets.

JEL codes: D83; G11; G14; G53; M41

Keywords: Generative AI; investors; information acquisition; information processing

1. Introduction

The rise of Generative Artificial Intelligence (GenAI), powered by advances in large language models (LLMs), has introduced numerous capabilities and opportunities to enhance investment decision-making for market participants.¹ A recent survey by Oliver Wyman (2024) shows that a growing share of respondents expresses interest in using GenAI for financial planning and advising, reaching 42% in 2024. GenAI is projected to become the leading source of investment advice for individuals in the near future, with 78% of small investors expected to adopt such tools by 2028 (Deloitte Report, 2024). Against this backdrop of rising attention and expectations, recent research has explored a variety of GenAI-assisted tasks related to stock investments, including processing corporate disclosures (Chang et al., 2025), assessing risks (Kim et al., 2025), forecasting firms' investment policies (Jha et al., 2025), and predicting stock price movements (Lopez-Lira and Tang, 2025).

Despite the hypothesized or documented benefits of GenAI in the stock market investment arena, there are at least three key reasons why investors may *not*, or *not fully*, utilize this new technology. First, investors may lack awareness of GenAI's capabilities, creating informational barriers that prevent them from accessing or exploiting its advantages. Second, even when aware, users may face resource constraints—such as limited time or cognitive capacity—that hinder their ability to select or effectively implement the tool's wide-ranging capabilities. Third, some investors may fundamentally distrust the quality and reliability of GenAI outputs (Glikson and Woolley, 2020), and therefore, when forming evaluations, assign them a lower weight relative to their own beliefs or to information from other sources. As such, it remains an open empirical question as to what extent market

¹ The Australian Securities and Investments Commission (ASIC) defines GenAI as “a type of AI that focuses on creating or generating novel content such as images, text, music, video, designs or recommendations” (ASIC, 2024).

participants, particularly retail investors, *actually* leverage GenAI when conducting stock-related research.

In this paper, we take a first step toward filling this gap by providing early-stage descriptive evidence on how retail investors interact with a GenAI platform for stock-related research. We focus on three dimensions: (1) users' information demand and the topics and tasks that they prompt, (2) GenAI responses and user-AI interaction dynamics, and (3) preliminary associations between user activity and market trading outcomes. Taken together, these analyses characterize the early adoption of GenAI tools by retail investors as reflected in their behavioral patterns, preferences, and heterogeneity, as well as the broader information environment in which these interactions occur.

Our analysis draws on detailed records of user queries, the corresponding GenAI answers, and additional user activity data, including user feedback and their behavior over time. These data are obtained from one of the largest GenAI platforms in China, serving more than 10 million monthly active users. This setting offers two key advantages for addressing our research question. First, although consumer-facing GenAI tools such as ChatGPT are widely accessible, researchers typically lack access to detailed data on user-platform interactions. By contrast, our partner platform provides back-end access to anonymized user data, offering a rare opportunity to observe investors' actual use of GenAI as it occurred. The data include both submitted queries and the corresponding answers across a wide range of topics and tasks, as well as user identifiers that allow us to track usage patterns within users over time. Second, China provides a particularly suitable environment for studying GenAI adoption among retail investors, who play a dominant role in the local stock market. This feature both enhances the representativeness of our sample, allowing us to observe a broad spectrum of GenAI interactions among typical retail market participants, and increases the relevance of our sample for aggregate stock market variables.

Starting from the universe of queries submitted on the platform during the first half of 2024, we first identify queries related to Chinese publicly listed companies using keyword searches for official firm names, including their common variants and abbreviations, in either the user queries or the corresponding GenAI answers. We then fine-tune a BERT model to classify and retain queries associated with stock market trading decisions. Next, we use a commercial LLM to categorize the resulting 1.7 million query-answer pairs along two dimensions: query topics and query tasks. We find that firm-specific queries span a broad range of informational dimensions, with the most frequent topics relating to firms' business outlook, production and operations, financial performance, and product and technology. For task classification, we follow the information processing framework of Blankespoor et al. (2020) and group queries into three primary stages—awareness, acquisition, and integration—along with more granular subtasks within each stage. Common subtasks include information search, impact analysis, news or event monitoring, and cause analysis, whereas trend prediction and summarization tasks occur relatively infrequently.

The first part of our analysis focuses on the information demand side and provides exploratory evidence on heterogeneity in investors' GenAI usage patterns, including the topics, tasks, frequency, and timing of queries around corporate information events. We begin by highlighting a key feature of our data: The sample period coincides with the early stage of the platform's public deployment, during which user engagement increases steadily from approximately 2,000 daily queries in early January to more than 8,000 by late June. A substantial share of queries is submitted outside trading hours and on weekends, indicating that the platform is used both for accessing timely market-related information and for less time-sensitive fundamental research.

User engagement is highly uneven. Approximately 40% of early adopters submit only a single stock-related query, whereas a small subset of users (8.5%) accounts for more intensive usage, submitting 20 or more queries. This pattern suggests that many users interact with the platform sporadically for their stock research needs, potentially driven by curiosity during the early stages of adoption or limited familiarity with GenAI tools. Such usage patterns are consistent with early adoption dynamics documented for other information technologies (Parthasarathy and Bhattacharjee, 1998; Bhattacharjee, 2001; Soliman and Tuunainen, 2022) and underscore the challenge GenAI providers face in fostering persistent engagement. At the same time, the nature of queried tasks evolves with continued interaction. Initial queries frequently focus on creating information awareness, but these decline over time, while information acquisition tasks become more prevalent. Information integration tasks start at, and remain at, relatively high levels throughout users' engagement, suggesting that some investors use early queries to overcome basic awareness frictions before transitioning to more targeted information needs.

Next, we examine cross-user and cross-day heterogeneity in query activity. We classify users with higher overall GenAI usage as relatively more engaged, and users who more frequently reference financial metrics as relatively more financially sophisticated. Firm-day-level analyses reveal systematic differences across these subgroups. More engaged and more sophisticated users are more likely to rely on GenAI for integration-oriented tasks, such as analyzing the causes and consequences of firm-specific developments and conducting financial statement analysis. In contrast, less engaged users more often use the platform for information awareness tasks.

We also document higher query activity on days with major corporate disclosures, including earnings announcements and management forecasts. These associations become less pronounced after controlling for contemporaneous media coverage, suggesting that query activity tends to parallel media attention rather than firm disclosures alone. In cross-sectional analyses, positive management forecasts are associated with higher query volume on average; however, this association is weaker for longer forecasts or forecasts that cover a broader range of topics. This pattern is consistent with more detailed management disclosures coinciding with a reduced need for additional contextual information from GenAI.

The second objective of our analysis is user-GenAI interactions, specifically how attributes of GenAI answers relate to immediate user feedback and subsequent engagement with the platform. To this end, we assess both the informational content of answers and their perceived usefulness. In particular, we construct a forward-looking signal alignment measure that captures whether the directional trading signal embedded in an answer is consistent with the stock's subsequent short-term market performance. We also examine several linguistic characteristics of the answers, including the prevalence of forward-looking language, subjectivity, and sentiment.

We find that the linguistic attributes of GenAI answers are strongly correlated with those of the corresponding user queries, indicating that responses partly mirror users' language choices. Building on these answer-level measures, we document that signal alignment is positively related to positive user feedback, including thumbs-up reactions, sharing, and copying. This pattern suggests that users are able to form judgments about the relevance and directional accuracy of trading signals shortly after reading an answer. Subjectivity and sentiment are also positively related to positive feedback, whereas answers that span multiple topics or exhibit a high degree of specificity are less likely to receive such reactions. By contrast, associations with negative feedback, measured by thumbs-down reactions, are generally lower in magnitude and statistical significance.

We next examine sustained engagement among users who return to the platform with stock market queries. For this group, signal alignment is positively associated with subsequent usage, consistent with users being more likely to re-engage following answers they perceive as informative. At the same time, continued engagement is negatively related to answer length and the inclusion of numerous external references, suggesting an aversion to overly lengthy or information-dense responses.

Finally, we track individual users' queries and answers over time to explore whether later queries reflect earlier interactions with GenAI. We find that, when early answers contain greater specificity or reference financial metrics, users are more likely to incorporate similar attributes or content in subsequent queries, even after controlling for the style and content of their initial queries. We also document that the appearance of three specific example financial metrics in early answers is associated with a higher likelihood that the user employs these same metrics in her later queries. Taken together, these findings indicate that users' subsequent information requests are systematically related to the content of their earlier interactions with GenAI.

In the final part of our exploratory analysis, we examine preliminary associations between GenAI usage and market-level trading variables. We document that higher query volume is associated with lower market liquidity, as reflected in a wider bid-ask spread and a higher Amihud illiquidity

measure.² At the same time, query activity is positively associated with abnormal trading volume and with proxies for informed trading. Taken together, these patterns suggest that periods of heightened GenAI usage coincide with greater trading activity and increased dispersion in information among market participants.

We conduct two cross-sectional analyses to further relate these associations to user interactions on the platform. First, the associations between query volume and trading outcomes are more pronounced on days when a larger share of answers receive positive user feedback, including thumbs-up reactions, sharing, or copying. Second, we find that the sentiment conveyed in GenAI answers is positively related to same-day abnormal returns, with stronger associations for answers that receive positive user feedback. These patterns are consistent either with users placing greater weight on answers they perceive as more useful or credible, or with GenAI answers reflecting contemporaneous market sentiment that is already incorporated into prices. While our design does not permit causal interpretations, the results indicate that user engagement with GenAI is systematically correlated with market trading variables.

Our study has both theoretical and practical implications. From a theoretical perspective, our primary contribution is the documentation of several novel facts about the information needs and information processing constraints of retail investors in the presence of GenAI. We show that investors' GenAI usage extends well beyond trading hours and follows a sequential information processing pattern consistent with the framework of Blankespoor et al. (2020). Although GenAI substantially lowers information acquisition and integration costs, users continue to rely heavily on other information channels—most notably the media—to become aware of corporate disclosures. Moreover, we find evidence consistent with managers' information provision substituting for investors' proactive information search via GenAI: Contemporaneous GenAI activity is lower when management forecasts provide richer contextual detail. This pattern suggests that GenAI is primarily used to fill informational gaps when disclosures are less complete, rather than to summarize already detailed information. These findings inform theories of investor attention and information processing by highlighting how new technologies interact with firms' disclosure choices.

Our empirical approach complements a growing body of literature that addresses the effects of GenAI on information processing and market outcomes using surveys (Blankespoor et al., 2026), experimental designs (e.g., Croom et al., 2026; Croom, 2026), or quasi-exogenous disruptions to GenAI

² In these tests, we control for information releases from intermediaries and investor activity on alternative channels, such as traditional search engines and investor-company interaction platforms. We caution that these associations cannot be attributed exclusively to GenAI usage, as we cannot fully disentangle GenAI-related activity from users' other information acquisition behavior.

access (Cheng et al., 2025; Bertomeu et al., 2025). While prior experimental work typically assigns narrowly defined tasks, such as processing earnings calls or corporate press releases, our evidence shows that retail investors also use GenAI extensively to explore broader, often nonfinancial information that supplements their existing information sets. In particular, users frequently engage in information search and in analyzing the causes and consequences of firm-specific developments. By observing actual user queries and responses on a large scale, our study reveals substantial heterogeneity both across users and across the types of information processing tasks they undertake. Likewise, this granularity provides a useful baseline for future research on the behavioral drivers, frictions, and downstream consequences of GenAI usage in financial decision-making.

Our findings also carry practical implications for regulators and GenAI service providers. From a regulatory perspective, there is increasing concern about whether retail investors can adequately assess the quality of GenAI trading recommendations and navigate the surrounding information environment.³ We document pronounced heterogeneity in GenAI usage patterns, attention allocation, and associated trading outcomes, suggesting that the benefits of GenAI are unevenly distributed across users. These disparities underscore the importance of continued regulatory oversight as GenAI tools become more deeply integrated into retail investing.

For GenAI service providers offering financial services or supporting capital market-related applications,⁴ our results highlight several design challenges and opportunities. First, nudging features such as push notifications or event-based alerts could help reduce users' awareness costs for relevant corporate information. Second, because users differ markedly in their ability to elicit useful information from GenAI, providing structured query templates or suggested prompts may improve average interaction efficiency, facilitate early learning, and help sustain usage. Finally, low retention rates may be mitigated by enhancing the usability of GenAI responses, for example, by delivering more accurate and concise answers or by pairing detailed outputs with clear executive summaries.

2. Literature Review and Research Setting

2.1 RETAIL INVESTORS' INFORMATION PROCESSING CONSTRAINTS AND NEW TECHNOLOGIES

Kim and Verrecchia (1994) provide theoretical support for the idea that public disclosures can widen information gaps among heterogeneous

³ A recent report by ASIC cautions that "AI models can provide information or advice that appears correct, but contains factual errors or fallacies," and highlights the risk of misleading or falsified information being presented to consumers (ASIC, 2024).

⁴ For example, Perplexity AI integrates the Quartr API into its search engine to provide real-time insights and AI-generated summaries of earnings calls for retail investors (Quartr, 2024).

investors in short windows around major disclosure events, when only a subset of investors is willing or able to bear the costs of processing new information. Building on this framework, Blankespoor et al. (2020) decompose information processing costs into three components: awareness costs, acquisition costs, and integration costs. Our paper focuses on small retail investors, who face substantial constraints throughout these sequential stages of information processing.

Prior research shows that retail investors frequently disregard value-relevant accounting information (Hirshleifer et al., 2008), even when awareness costs are meaningfully reduced (Blankespoor et al., 2019). As corporate disclosures have become longer and more complex over time (Dyer et al., 2017), many retail investors may lack the cognitive capacity or time required to locate, extract, and integrate relevant information. Consequently, they tend to invest in firms with clearer and more concise disclosures (Lawrence, 2013) or to process disclosures in a more superficial manner than institutional investors (Veldkamp, 2011). Consistent with these constraints, empirical evidence suggests that technological advances in information dissemination have, at best, delivered limited benefits to small investors. In particular, retail investors are slower to adapt to, and hence benefit from, mandatory disclosure technologies such as XBRL and EDGAR (Blankespoor et al., 2014; Gomez, 2024).

More recent studies indicate that platforms that either integrate rich information cues (Chau, 2025) or facilitate direct communication between investors and firms (Lee and Zhong, 2022) can help self-directed retail investors make more informed decisions. Notably, several studies document that retail investors prefer processed or summarized information from intermediaries over raw accounting data or primary disclosures (Lawrence et al., 2017; Blankespoor et al., 2019; Chau, 2025). This preference is somewhat surprising, as processed information often entails delayed access and reduced information granularity. For example, Lee and Zhong (2022) show that firms listed on the Shenzhen (Shanghai) Stock Exchange take, on average, seven (22) days to respond to investor questions and respond to only 93% (77%) of inquiries. Such long delays limit the usefulness of these channels for time-sensitive information.

In contrast, GenAI platforms have the potential to lower awareness and acquisition costs while delivering relevant information much more promptly. More broadly, prior research suggests that retail investors are interested in fundamental information but are often unwilling or unable to process it directly from filings and disclosures. In this context, new information processing technologies are likely to benefit retail investors only if they (1) substantially reduce monetary and cognitive costs and (2) meaningfully alleviate information integration frictions.

2.2 CAPABILITIES OF LLMs IN STOCK MARKET INVESTMENT DECISIONS

While financial service providers have long adopted AI in data processing and decision-making (e.g., Liu et al., 2023; Coleman et al., 2022), the

release of LLMs such as GPT-4 has substantially broadened access to GenAI tools for stock market participants, including retail investors. Since then, the problem-solving performance of these models in financial contexts has improved considerably, as illustrated by higher CFA exam pass rates of GPT-4 compared to GPT-3.5 (Callanan et al., 2023).

Research highlights three key roles of GenAI in investor decision-making. First, GenAI can enhance information awareness by substituting for traditional intermediaries, offering more timely and higher quality summaries of disclosures.⁵ Second, GenAI facilitates information acquisition by enabling users to efficiently extract and tailor firm-specific information from diverse sources. For example, GPT models can automate data extraction from corporate reports (Gupta, 2023; Ni et al., 2023), capture firm-specific risks related to innovation (Kim et al., 2025), and uncover content in filings that is not easily identifiable with traditional methods (Breitung and Müller, 2025; Costello et al., 2025). Third, GenAI assists with information integration: It can connect textual and numerical disclosures to improve forecasting of cash flows and returns (Kim and Nikolaev, 2024), while GPT-based investment scores have been shown to predict abnormal returns (Jha et al., 2025).⁶

Rather than relying passively on third-party summaries, retail investors increasingly use GenAI to interact directly with financial information in a personalized way (Blankespoor et al., 2026). This shift enables proactive discovery of disclosures and offers new insights into investors' information needs across awareness, acquisition, and integration stages. Our paper analyzes query-level data to shed light on the topics investors focus on and the tasks they assign to GenAI. We further link user feedback and query activity to stock market outcomes, providing evidence on how AI-delivered information is incorporated into trading decisions.

2.3 RESEARCH SETTING

Our research setting and data source are a conversational GenAI platform developed by a leading technology company in China. Launched in mid-2023, the platform has rapidly evolved to handle queries across a broad range of subject domains, including finance, business, and technology. Its user-friendly interface and interaction design closely resemble those of mainstream GenAI platforms such as ChatGPT and Perplexity (see online appendix OA.1 for a mock-up of the platform interface), which enhances the relevance and potential generalizability of our findings to other platforms.

⁵ For example, Alpha, an AI-powered mobile app, provides real-time market insights (see: <https://www.prnewswire.com/news-releases/alpha-a-new-ai-watchlist-app-launches-to-give-individual-investors-real-time-insights-on-the-markets-302330059.html>).

⁶ At the macroeconomic level, Woodhouse and Charlesworth (2023) use LLMs to analyze central bank speeches to forecast monetary policy decisions.

The GenAI system is connected to the internet, enabling access to real-time information. It operates under a freemium model: Standard search and chat interactions via the web and mobile app interfaces are freely available to all users. Unregistered users can access only the basic AI search functions and are limited to a small number of queries per day (typically four to five); registered users must verify their identity by linking a mobile phone number to gain free access to the full range of features, including natural language chats and enhanced interaction tools (e.g., uploading documents, generating slides). These freely available interactions constitute the source of our data set. In contrast, API-based access for enterprises is subject to a fee, and these interactions are not in our data set.

By April 2024, the platform ranked among the top GenAI platforms in China by user adoption, with over one (10) million daily (monthly) active users. Its user base spans academia, business, and the general public. The platform's rapid growth reflects rising demand for GenAI tools in a broad range of everyday information-seeking and decision-support activities. Importantly for our setting, its scale allows us to filter and analyze a sufficiently large subsample of user queries specifically related to stock market investment.

Our institutional setting has several important features. First, China's stock market is dominated by retail investors, who account for nearly 80% of daily trading volume (Jones et al., 2025). Many of these investors have relatively low levels of formal education and financial literacy (Titman et al., 2022), which can pose challenges for independently processing and integrating complex investment-related information. Second, GenAI adoption in China is notably high. Survey evidence indicates that 83% of Chinese respondents report using GenAI, compared with 65% in the United States and a global average of 54% (Reuters, 2024). At the same time, major foreign GenAI services, such as OpenAI's ChatGPT, are not officially accessible in China. This restriction has led users to rely primarily on domestic GenAI providers, including the platform providing the data for this paper.

2.4 DATA AND SAMPLE CONSTRUCTION

2.4.1. Sample Selection. We analyze a comprehensive data set of user queries and corresponding GenAI answers from the platform during the first half of 2024. These data provide detailed information on user behavior, query content, and user-platform interactions, offering a unique opportunity to examine how market participants leverage GenAI tools for stock-related research and decision-making.

We begin with the universe of 478,153,244 non-blank, valid query-answer pairs submitted between January 1 and June 30, 2024. We augment this data set with user interaction data, including direct feedback on GenAI answers—specifically, whether users provide thumbs-up or thumbs-down reactions, share answers via platform-generated links, or copy answer content for potential use outside the platform. Our analysis focuses on the Chinese stock market, the primary investment universe for Chinese retail investors,

and includes all A-share companies listed on the Shanghai and Shenzhen Stock Exchanges during fiscal year 2023.

We retain only those query-answer pairs in which either the query or the corresponding answer contains an identifiable reference to at least one listed company, based on official company names or commonly used name variants or abbreviations.⁷ This procedure yields a data set that comprehensively captures firm-referenced query-answer interactions. To reduce noise arising from incorrect classifications, we apply a fine-tuned Chinese BERT model to distinguish stock-market-related queries from unrelated queries, particularly in cases where company names overlap with generic terms (e.g., “Robots”). After taking these filtering steps, our final sample consists of 1,746,075 stock-market-related query-answer pairs referring to 5,278 unique firms. Online appendix OA.2 provides illustrative examples from this sample.

We further classify these query-answer pairs based on query specificity. A total of 1,449,174 queries (83%) are *firm-specific*, explicitly referencing at least one individual company, while the remaining 296,901 queries (17%) are *general* in nature and pertain to broader market- or industry-level topics. In the latter case, references to specific firms are typically introduced by the GenAI platform in its responses. This classification is based on manual labeling of a training subsample followed by automated classification using a Chinese BERT model.

After matching each query-answer pair to the relevant firm or firms,⁸ we construct several data sets tailored to our empirical analyses. First, we aggregate the data to the *firm level* for cross-sectional analyses of firm characteristics. Second, we aggregate query-answer pairs to the *firm-day level* for event-day analyses. Third, we construct a *user-query-level* data set by linking query attributes and user characteristics using anonymized user and session identifiers provided by the platform.⁹ This data set allows us to track individual users’ behavior over time and to examine immediate feedback on each answer. Finally, we construct a *user-level* data set to study how users’ early queries and answers relate to their subsequent queries. Financial and stock market data used in the analyses are obtained from the China Stock Market & Accounting Research (CSMAR) database. Appendix A contains the definitions of the main variables and table 1 reports descriptive statistics.

⁷ In the Chinese stock market, listed companies are frequently referenced by standardized abbreviations, analogous to ticker symbols in the United States. For example, “Kweichow Moutai Co. Ltd.” is commonly referred to as “Moutai.” In addition, the platform’s built-in LLM normalizes and auto-corrects company names, even when users employ nonstandard references, which helps to reduce false negatives arising from inconsistent naming.

⁸ We use a commercial LLM to improve firm matching and identify company names and stock codes in user queries and responses. In addition, we construct a mapping table linking stock codes, official company names, and common abbreviations, and verify matches using this crosswalk table.

⁹ A session corresponds to a single conversation thread, analogous to a chat session in ChatGPT.

TABLE 1
Descriptive Statistics for Regression Analyses

Variables	N	Mean	SD	p25	p50	p75
Firm-day-level variables:						
#Specific_Queries	596,989	1.416	12.185	0.000	0.000	1.000
EA_Pos	596,989	0.005	0.073	0.000	0.000	0.000
EA_Neg	596,989	0.005	0.067	0.000	0.000	0.000
MF_Pos	596,989	0.002	0.044	0.000	0.000	0.000
MF_Neg	596,989	0.003	0.050	0.000	0.000	0.000
Event_Other	596,989	0.137	0.344	0.000	0.000	0.000
AnaReport	596,989	0.020	0.140	0.000	0.000	0.000
Media	596,989	0.159	0.365	0.000	0.000	0.000
Log(1+SVI)	596,989	5.055	2.926	5.037	6.292	6.983
Log(1+Retail Platform)	596,989	0.199	0.409	0.000	0.000	0.000
PosFBRatio	150,333	0.087	0.224	0.000	0.000	0.000
Avg_Sentiment	142,532	0.247	0.317	0.000	0.250	0.452
Avg_Sentiment_PosFB	22,714	0.309	0.355	0.075	0.333	0.556
Avg_Sentiment_NonPosFB	22,714	0.245	0.271	0.093	0.258	0.404
AbnSpread	594,700	0.317	3.725	−1.487	−0.114	1.515
Amihud Illiquidity	596,989	4.412	6.132	0.752	2.196	5.438
AbnVol	595,553	−0.065	2.216	−0.698	−0.151	0.308
VPIN	596,989	17.809	5.226	14.100	17.362	21.029
AbnReturn	594,844	−0.043	2.860	−1.425	−0.095	1.261
Log(MV)	596,989	15.594	1.075	14.834	15.372	16.169
Answer-level variables:						
Positive_FB (×100)	376,491	13.514	34.187	0.000	0.000	0.000
Thumbs_Down (×100)	376,491	0.096	3.104	0.000	0.000	0.000
#Future_Queries	376,491	84.679	514.998	2.000	9.000	29.000
Signal_Align (Future 7 days)	376,491	0.365	0.477	0.000	0.000	1.000
Forward_Looking	376,491	0.376	0.293	0.125	0.333	0.585
Subjectivity	376,491	0.413	0.252	0.231	0.400	0.585
Sentiment	376,491	0.335	0.428	0.045	0.375	0.667
Fog	376,491	10.698	2.867	8.750	10.255	12.125
Topic_Diversity	376,491	0.148	0.113	0.048	0.136	0.227
Specificity	376,491	0.194	0.132	0.095	0.168	0.267
#References	376,491	5.483	8.621	0.000	3.000	6.000
#Words	376,491	556.125	473.234	288.000	434.000	633.000
User-level variables:						
Specificity_First3_Queries	78,453	0.320	0.202	0.157	0.287	0.445
Specificity_First3_Answers	78,414	0.182	0.109	0.101	0.165	0.244
Specificity_Post3_Queries	78,887	0.315	0.192	0.171	0.288	0.417
Any_FM_First3_Queries	78,887	0.139	0.273	0.000	0.000	0.333
Any_FM_First3_Answers	78,887	0.536	0.356	0.333	0.667	1.000
Any_FM_Post3_Queries	78,887	0.160	0.276	0.000	0.000	0.214
ROE_First3_Queries (×100)	78,887	0.700	8.336	0.000	0.000	0.000
ROE_First3_Answers (×100)	78,887	3.180	17.548	0.000	0.000	0.000
ROE_Post3_Queries (×100)	78,887	3.201	17.602	0.000	0.000	0.000
Solvency_First3_Queries (×100)	78,887	0.956	9.730	0.000	0.000	0.000
Solvency_First3_Answers (×100)	78,887	1.605	12.566	0.000	0.000	0.000
Solvency_Post3_Queries (×100)	78,887	3.130	17.412	0.000	0.000	0.000
Accounts_Receivable_First3_Queries (×100)	78,887	0.838	9.115	0.000	0.000	0.000
Accounts_Receivable_First3_Answers (×100)	78,887	3.172	17.524	0.000	0.000	0.000
Accounts_Receivable_Post3_Queries (×100)	78,887	2.709	16.234	0.000	0.000	0.000

This table presents descriptive statistics for variables used in subsequent regression analyses. All variables are defined in appendix A.

2.4.2. Classification of Query Topics. We identify query topics using a three-step procedure. First, we apply Latent Dirichlet Allocation (LDA) topic modeling to identify the latent topics of queries and their corresponding answers. This unsupervised machine learning method allows topics to emerge from the data without requiring prior assumptions.¹⁰ Second, based on a manual review of the most prevalent LDA topics and their most distinctive keywords, we define a set of stable and interpretable topic categories. Third, we use a commercial LLM to classify all query-answer pairs into these refined categories.¹¹

Importantly, we do not require the topics to be mutually exclusive. On the one hand, the boundaries between certain topics are subjective and can be ambiguous—for instance, a query about the market landscape of a new product may arguably fall under both the “Competition” and “Product and Technology” topics. On the other hand, and more fundamentally, some queries explicitly address multiple topics simultaneously, for example, by posing several sub-questions within a single prompt.

2.4.3. Classification of Query Tasks. Although prior literature has explored various types of tasks that GenAI could perform in principle, we aim to provide confirmatory evidence on whether and how these tasks are actually implemented by investors in practice. We start by manually reviewing a random sample of 20,000 queries to identify task types that are both frequently observed and conceptually distinct; the resulting task list and representative examples are provided in appendix C. We then apply a commercial LLM to classify all 1,746,075 sample queries based on this task taxonomy.

We then broadly classify queries into tasks related to information processing, as well as other types of tasks. For information processing, we further break down tasks into more granular categories along the dimensions of information awareness, acquisition, and integration, following the framework of Blankespoor et al. (2020). The other task categories include *generative tasks* (e.g., drafting reports or slides) and *educational tasks* (e.g., learning about financial or stock market concepts). As with the topic classification, we do not impose a mutual exclusivity constraint—individual queries may span multiple task (sub)categories.

3. GenAI Usage Patterns: Intensity, Purposes, and Timing

In this section, we present descriptive evidence on how users engage with GenAI for stock market research. We document overall usage intensity, the

¹⁰ We evaluate multiple LDA model specifications by varying the number of topics. For our evaluation and selection of the best-fitting structure, we use both perplexity scores (Blei et al., 2003) and the OpTop metric (Lewis and Grossetti, 2022).

¹¹ See appendix B for a full list of topics. Detailed workflow charts and prompt templates are provided in online appendix OA.3 and OA.4; online appendix OA.5 compares the topics classification from this LDA-initiated approach with that from an entirely LLM-based approach.

purposes for which users employ GenAI, as captured through the topics and tasks embedded in their queries, heterogeneity across users, and the timing of query activity around corporate information events. Together, these descriptive patterns shed light on stock market participants' early-stage adoption of GenAI and their underlying information needs.

3.1 OVER-TIME AND GEOGRAPHIC VARIATIONS IN OVERALL PLATFORM USAGE

In online appendix OA.6, we present exploratory graphical evidence on platform usage over the sample period and across geographic regions. From January to June 2024, both query volume and the number of unique users increase steadily, with growth in query volume broadly tracking the expansion of the user base. Usage also varies geographically. Adoption is highest in the eastern coastal regions of China, which host major financial and economic hubs, and substantially lower in inland regions. Finally, within-week patterns indicate that query activity peaks from Monday to Thursday and declines modestly over the weekend, suggesting that users' interest in stock market research extends beyond formal trading days.

Figure 1 illustrates the density of queries in half-hour intervals over an average trading day. The figure shows a pronounced concentration of activity during trading hours, which are separated by dashed vertical lines and highlighted with a light blue background, particularly between 9:30 a.m. and 4:00 p.m. Query activity increases sharply prior to market opening, peaks around market close, and gradually declines thereafter. In addition, we observe a smaller peak around 10:00 p.m., which may reflect users gathering information after work or before the end of the day. Overall, this temporal pattern suggests that the platform serves as an information source for both real-time market monitoring and after-hours investment research.

Focusing on identifiable individual GenAI users, defined as those registered with unique user IDs, figure 2 plots the distribution of per-user query counts for users who submitted their first stock-market-related query in January 2024.¹² The distribution is highly skewed. More than half of these users submit only one or two queries, and the number of users declines sharply as query counts increase. Only 8.5% of users submit 20 or more queries. This pattern indicates that most users interact with the platform only sporadically, while sustained engagement is concentrated among a relatively small subset of users. Such sporadic usage is consistent with early adoption patterns of new information technologies, where many users disengage after

¹² We focus on early entrants to mitigate truncation bias at the end of the sample period. Among users who joined the platform between February and June, the share of users submitting only one query during the sample period increases mechanically over time, reaching 65.5% for the June cohort.

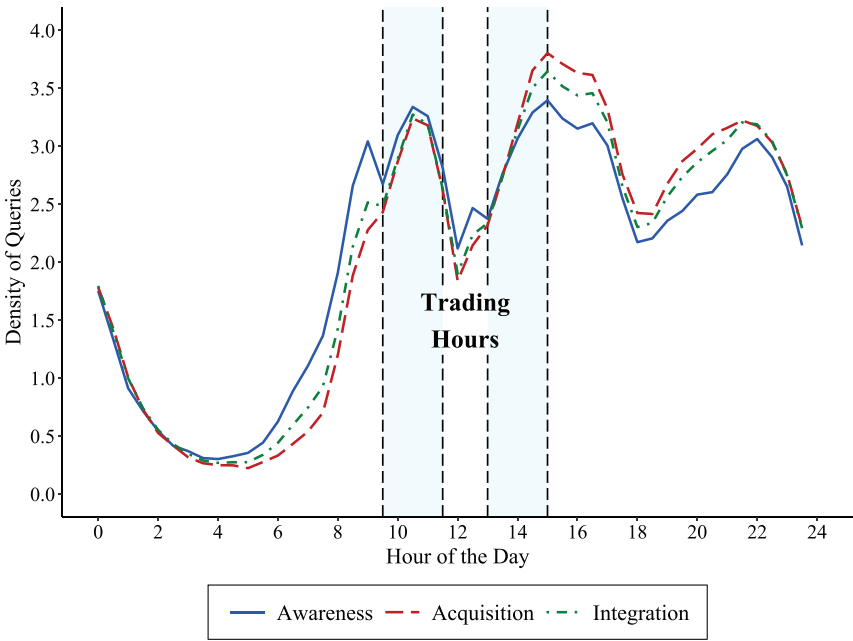


FIG. 1.—Density plots of queries across (trading) hours. This figure presents the average density of stock-related queries for each half-hour interval (e.g., 3:00 p.m.-3:30 p.m. is represented as 15 on the 24-hour clock) across trading days. Trading hours are highlighted with a light blue background. The blue solid line represents the density of information awareness tasks, the red dashed line indicates the density of information acquisition tasks, and the green dot-dashed line corresponds to the density of information integration tasks.

a limited number of interactions (Parthasarathy and Bhattacharjee, 1998; Soliman and Tuunainen, 2022).¹³

Taken together, these descriptive patterns point to two salient aspects of platform usage. First, query activity reflects market-related information needs with varying time sensitivity, showing higher intensity during trading hours in addition to continued use during evenings and weekends. Second, most users interact with the platform sporadically and in an exploratory manner, while a much smaller subset exhibits sustained usage over time.

3.2 TOPICS OF QUERIES AND ANSWERS

Based on all topics identified from users’ interactions with the platform, appendix B reports the top 10 query topics, the top 15 words associated with each topic, and the corresponding topics mentioned in the GenAI answers. The six topics most frequently queried are business outlook (26.32%

¹³ Although most users submit fewer than five firm-specific queries, they may still use the platform for other purposes beyond stock-related research. Online appendix OA.7 summarizes additional non-investment-related tasks observed on the platform.

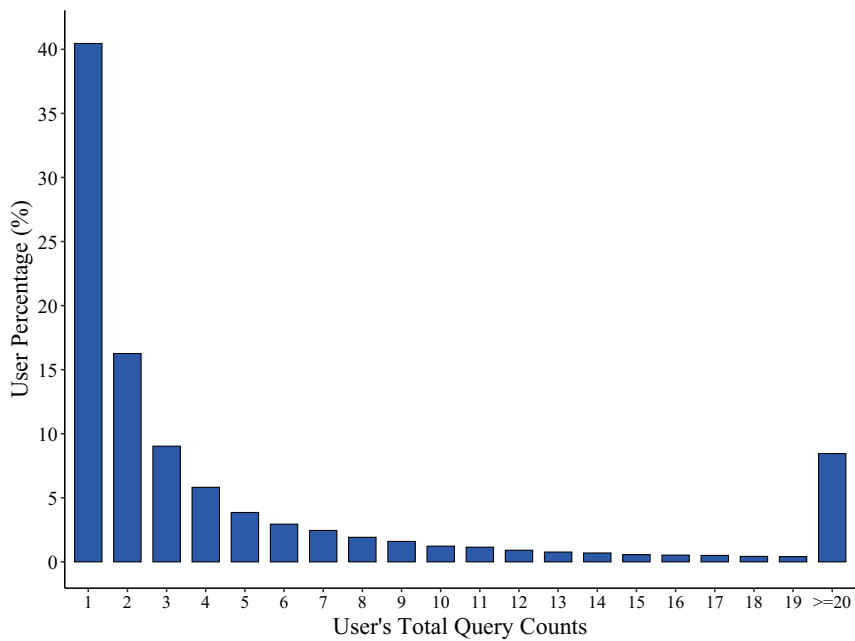


FIG. 2.—Density plot of per-user queries. This figure displays the density distribution of users by their query count. The x -axis represents the total number of queries submitted by a user, while the y -axis shows the density of users for each query count. To account for potential truncation bias due to the sample end date, we restrict the sample to users who began using the platform in the first month of our sample period (i.e., January 2024). Users without registered user IDs are excluded, as their query sequence cannot be tracked.

of all sample queries), production and operations (25.56%), financial performance (23.26%), product and technology (22.26%), stock market performance (19.60%), and competition (18.04%). By contrast, the remaining four query topics are far less frequent, with incidence rates ranging from 6.23% to 7.43%.

In the second column, we report the most frequent unique words used under each query topic. User queries predominantly focus on firms’ core business activities and financial positions, as reflected in frequently used terms such as “development prospects,” “profit forecast,” and “operating capacity” under topics including “Business Outlook,” “Production and Operations,” and “Financial Performance.” Topics related to “Product and Technology” emphasize interest in innovation, with common terms such as “R&D” and “technology,” whereas queries related to “Stock Market Performance” and “Competition” commonly reference words such as “stock price,” “competitor,” and “market share.” Overall, these patterns suggest that users employ GenAI to collect and synthesize information across a broad set of strategic, operational, and market-related dimensions.

Answer topics generally align with the topic of the corresponding query. At the same time, GenAI frequently supplements its responses by

incorporating information from related topics. For instance, answers to queries about financial performance often include discussion of production and business operations. This tendency suggests that GenAI provides contextual information that may help users interpret numerical facts and financial metrics in more complex settings.

3.3 TASKS OF QUERIES

We present the distribution of query tasks, along with their frequencies and examples, in appendix C. The majority of queries fall into information integration tasks (77.6%), followed by acquisition tasks (44.5%), and awareness tasks (38.8%). This pattern suggests that retail investors are particularly inclined to outsource the entire process of information processing to GenAI, by directly seeking integrated investment signals rather than intermediate inputs.

Among the subtasks across all three task categories, information search is the most frequent, appearing in nearly one-third of all queries (32.7%). Investors also frequently inquire about the causes and impacts of specific corporate decisions or market phenomena (16.7% and 18.7%, respectively), which we categorize as integration tasks. Such queries help users contextualize or interpret new developments or events. News or event monitoring is the third most-often requested task, with 17.7% of queries, which we categorize as raising investor awareness. Interestingly, despite GenAI's demonstrated ability to summarize financial statements and highlight key facts (Kim et al., 2025), summarization tasks account for only 3.3% of all queries in our sample. This highlights an opportunity for further educating retail investors on the potential value of using GenAI for summarization and structured financial statement analysis.

Figure 3 presents a user-level view of how the composition of information processing tasks evolves over a user's sequence of queries. Integration-related tasks consistently account for the largest share of queries throughout the interaction history. Nevertheless, platform engagement is often initiated with awareness-related queries, such as requests for stock recommendations or monitoring corporate disclosures. The likelihood of submitting an awareness-related query exceeds 50% for first-time queries but declines steadily with continued platform use. In contrast, the probability of submitting acquisition-related tasks increases over subsequent interactions. We interpret these dynamics as reflecting a gradual shift from creating general awareness toward more targeted information needs as users gain experience with the platform. Integration tasks remain persistently prevalent, fluctuating within a relatively narrow range from approximately 72.1% to 80.6%, suggesting that users frequently rely on GenAI to perform complex reasoning and synthesis throughout their platform uses. Taken together, these descriptive results suggest that initial queries tend to reflect broader and less structured informational demands, consistent with awareness frictions, while subsequent interactions increasingly focus on targeted information acquisition and processing.

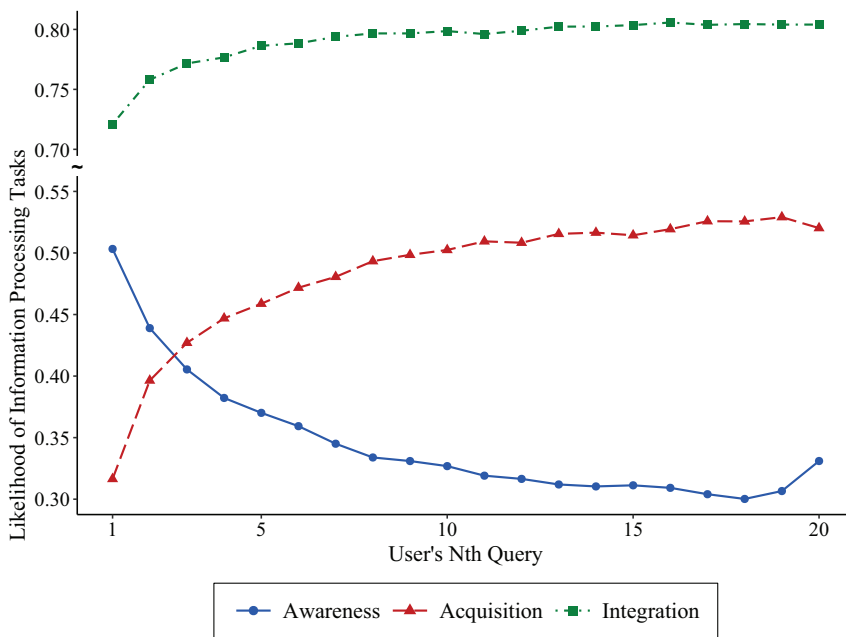


FIG. 3.—Distribution of information processing tasks in query sequence. This figure presents the distributions of user queries across information awareness, acquisition, and integration tasks, by the user's N th query. N refers to the sequential order of a given user's queries. The query sequence is limited to 20 queries, whereby we condense all queries beyond the 20th into the 20th query. The blue solid line represents the density of information awareness tasks, the red dashed line indicates the density of information acquisition tasks, and the green dot-dashed line corresponds to the density of information integration tasks. A break is introduced in the y-axis between 0.55 and 0.70 to accommodate the higher frequency of integration tasks, while preserving visibility for the other task types. Users without registered user ID are excluded, as their query sequence cannot be tracked.

3.4 USER HETEROGENEITY

Although the majority of users in our sample are retail investors who are relatively new to GenAI and interact with it primarily for exploratory purposes, a subset engages with the technology more intensively and in a more sophisticated manner. We capture user engagement using persistent AI usage, motivated by the idea that more efficient use of GenAI generates informational benefits that, in turn, reinforces the incentives for continued engagement. Users in the top decile of total queries within their month-of-entry cohort are classified as “High” AI usage, with the remainder classified as “Low” AI usage.¹⁴ We separately proxy for user sophistication based on

¹⁴We benchmark users relative to their month-of-entry cohort to mitigate truncation bias arising from the fixed end date of the sample. Partitioning at the top decile is motivated by the highly skewed distribution of query activity (figure 2), in which a small fraction of users accounts for a disproportionate share of total queries.

the frequent use of financial metrics in queries, which suggests familiarity with financial statement analysis. Users in the top decile of average financial metrics mentioned per query are classified as “High” metrics usage and the remainder as “Low” metrics usage.

Table 2 summarizes usage patterns across these user partitions. More engaged and financially sophisticated users both disproportionately focus on financial topics and financial statement analysis tasks, as well as information acquisition and integration activities such as searching, validating information, and analyzing causes and impacts. In contrast, less engaged and less sophisticated users are more likely to use GenAI for information awareness tasks and to ask broader, less structured questions about firm prospects.

These differences in usage patterns suggest that more engaged and more financially sophisticated users employ GenAI in a more targeted and analytical manner, whereas other users tend to rely on the tool more broadly for exploratory guidance. Consistent with this interpretation, supplementary analyses indicate that tasks more frequently undertaken by engaged and sophisticated users, such as financial statement analysis and summarization, are also associated with stronger simulated trading performance based on signals extracted from GenAI answers.¹⁵

3.5 INDUSTRY ATTENTION AND FIRM CHARACTERISTICS

Online appendix OA.9 examines variation in GenAI query intensity across industries and firms. In brief, the Automobile Manufacturing sector accounts for the largest share of queries (11.67%), followed by Electrical Machinery and Equipment Manufacturing (9.90%) and Computer, Communication, and Other Electronic Equipment Manufacturing (8.68%). Banking and Alcohol also attract substantial attention on a per-firm basis, likely reflecting their macroeconomic importance and, in the case of Alcohol, the presence of widely followed “star” firms.¹⁶ Overall, the most frequently queried industries tend to be those associated with innovation, high growth potential, or strong consumer visibility.

We also analyze the relation between firm-level characteristics and GenAI query activity, both in the full sample and across partitions by query topics and tasks (results tabulated in online appendix OA.10). Larger firms, firms with stronger recent performance, and firms with higher market-to-book ratios receive more queries, whereas institutional ownership is negatively

¹⁵ In online appendix OA.8, we backtest hypothetical trading strategies that follow GenAI signals prior to earnings announcements, both for all queries and for the query cross-sections of tasks, topics, and user feedback types. Queries related to financial statement analysis tasks and financial performance topics—predominantly submitted by more engaged and sophisticated users—are associated with higher simulated abnormal returns, whereas generic or awareness-oriented queries are not.

¹⁶ In China, firms such as Kweichow Moutai and Wuliangye—leading producers of premium distilled spirits—are commonly viewed as “star stocks” and are especially popular among retail investors due to strong brand recognition and high historical returns.

TABLE 2
User-Based Cross-Sections of Query Topics and Tasks

	Query-Level Subsamples by Users' AI Usage			Query-Level Subsamples by Users' Financial Metrics Usage		
	(1) All Registered Users	(2) High (#Users = 36,612) (#Queries = 974,082)	(3) Low (#Users = 318,241) (#Queries = 644,344)	(4) High (#Users=38,637) (#Queries=364,051)	(5) Low (#Users=316,216) (#Queries=1,254,375)	Diff. (4) – (5)
Query Topics:						
<i>Financial</i>	0.279	0.340	0.171	0.168 ^{***}	0.156	0.495 ^{***}
<i>Business_Outlook</i>	0.318	0.322	0.311	0.012 ^{***}	0.320	–0.008 ^{***}
<i>Production</i>	0.307	0.339	0.249	0.091 ^{***}	0.255	0.207 ^{***}
<i>Stock_Market</i>	0.237	0.230	0.249	–0.019 ^{***}	0.250	–0.052 ^{***}
<i>Product_Tech</i>	0.270	0.255	0.298	–0.043 ^{***}	0.301	–0.192 ^{***}
<i>Competition</i>	0.219	0.214	0.228	–0.014 ^{***}	0.226	–0.026 ^{***}
Query Tasks:						
<i>Info_Aware</i>	0.388	0.323	0.486	–0.163 ^{***}	0.464	–0.337 ^{***}
<i>Info_Search</i>	0.327	0.376	0.252	0.124 ^{***}	0.286	0.182 ^{***}
<i>Cause_Analyze</i>	0.168	0.196	0.125	0.071 ^{***}	0.149	0.084 ^{***}
<i>Summarize</i>	0.033	0.040	0.023	0.017 ^{***}	0.020	0.062 ^{***}
<i>Compare</i>	0.136	0.148	0.119	0.029 ^{***}	0.119	0.076 ^{***}
<i>Trend_Predict</i>	0.136	0.114	0.168	–0.054 ^{***}	0.150	–0.064 ^{***}
<i>Impact_Analyze</i>	0.187	0.215	0.145	0.071 ^{***}	0.172	0.068 ^{***}
<i>Overall_Evaluate</i>	0.158	0.138	0.187	–0.048 ^{***}	0.176	–0.080 ^{***}
<i>FS_Analyze</i>	0.146	0.196	0.070	0.126 ^{***}	0.068	0.346 ^{***}

This table reports heterogeneity in GenAI queries across user subgroups. The sample is restricted to the 1,618,426 queries from 354,853 registered users, further partitioned along two user dimensions: (i) AI usage, where those in the top decile of total queries within their month-of-entry user cohort are classified as “High” AI usage and the remainder as “Low” AI usage; and (ii) financial metrics usage, where those in the top decile of average financial metrics per query across the full distribution are classified as “High” and the remainder as “Low.” For each subgroup pair, we report the share of queries by topic and by task, along with differences in means. Differences significant at the 1% level are marked with ***.

associated with query activity, consistent with the predominance of retail investors on the platform. Leverage is positively related to queries concerning financial position and production, and firms audited by Big 4 auditors or receiving greater media coverage also attract more attention. When queries are partitioned by task type, leverage is most strongly associated with financial statement analysis, while firms with higher fixed assets receive fewer information integration queries, such as those involving trend predictions. Firms with higher R&D spending attract more queries related to competition, products, and technology, as well as comparative analysis tasks, suggesting that users rely on GenAI to benchmark innovation-intensive firms.

Overall, these findings indicate that users' information-seeking behavior varies systematically with firm size, performance, risk profile, and growth orientation, and that these patterns differ meaningfully across query topics and task types.

3.6 FIRM-SPECIFIC QUERIES AROUND CORPORATE INFORMATION EVENTS

Motivated by prior evidence that investors' information acquisition intensifies around corporate disclosure events (Drake et al., 2012; Lee and Zhong, 2022), we examine how firm-specific GenAI queries vary in relation to corporate information events. We estimate the following regression using firm-day observations:

$$\begin{aligned} \#Specific_Queries_{i,t} = & \beta_1 EA_Pos_{i,t} + \beta_2 EA_Neg_{i,t} + \beta_3 MF_Pos_{i,t} \\ & + \beta_4 MF_Neg_{i,t} + \beta_5 Event_Other_{i,t} + \beta_6 AnaReport_{i,t} \\ & + \beta_7 Media_{i,t} + \text{Firm FE}_i + \text{Date FE}_t + \varepsilon_{i,t}, \end{aligned} \quad (1)$$

where $\#Specific_Queries_{i,t}$ denotes the raw count of queries specifically related to firm i on day t . The independent variables include separate indicator variables for five types of corporate information events: positive earnings announcements (EA_Pos), negative earnings announcements (EA_Neg), positive management forecasts (MF_Pos), negative management forecasts (MF_Neg), and other unsigned material events ($Event_Other$), such as mergers and acquisitions.¹⁷ We classify earnings announcements as positive or negative based on the sign of unexpected earnings, defined as the difference between current-quarter earnings and earnings from the same quarter in the prior year. An announcement is categorized as positive ($EA_Pos = 1$) if unexpected earnings are above

¹⁷We define earnings announcements as the days on which the public first learns about a firm's quarterly earnings through a formal disclosure. Accordingly, we rely on the more commonly observed mandatory filings of periodic reports rather than the much less frequent voluntary pre-filing earnings announcements to identify such events. This definition is empirically consistent with Lee and Zhong (2022), who report an incidence rate of earnings announcements of 1.26% of firm-days. In our sample and sample period, the corresponding rate is 0.99%. By contrast, voluntary pre-filing earnings disclosures occur on only 0.11% of firm-days, and we find no evidence of a notable increase in query activity on those days.

zero, and negative ($EA_Neg = 1$) otherwise. For management forecasts, we exploit a regulatory feature in China that requires firms to explicitly label forecasts as indicating either an increase ($MF_Pos = 1$) or a decrease ($MF_Neg = 1$) in net profits.¹⁸ Firm-days with information releases by intermediaries are identified using separate indicator variables for analyst report releases (*AnaReport*) and for news articles (*Media*). All specifications include firm fixed effects and date fixed effects to control for time-invariant firm characteristics and common time-specific factors.

Table 3 reports the results. Column 1 shows that firm-specific queries increase significantly around corporate disclosures. Both positive and negative earnings announcements are associated with higher query volume, with economically larger effects following negative earnings news. Query activity also rises on days with other material corporate events. However, once we account for contemporaneous information releases by intermediaries in column 2, the coefficients on earnings announcements become statistically insignificant, while media coverage remains strongly associated with increased query activity. This pattern suggests that users' information demand on the GenAI platform is more closely aligned with the dissemination and interpretation of corporate events by intermediaries than with firms' original disclosures alone.

Columns 3 and 4 further examine heterogeneity in user activity on days with management forecast issuance by interacting forecast indicators with measures of disclosure informativeness. Specifically, *More_Topics* is an indicator equal to one if a management forecast discusses a number of distinct topics above the sample median, and *Longer* is an indicator equal to one if the forecast's word count exceeds the sample median.¹⁹

The results indicate that the positive association between favorable management forecasts and firm-specific query volume is weaker when forecasts are longer or cover more topics. In contrast, relatively brief or narrowly focused positive forecasts are associated with higher query activity. These patterns are consistent with a substitution mechanism in information acquisition, whereby more informative management forecasts are associated with reduced demand for additional contextual information on the GenAI platform. One possible interpretation is that qualitative statements accompanying favorable earnings news may be perceived as

¹⁸ Management forecasts in China fall under a hybrid disclosure regime: While issuance is voluntary in principle, regulations by the China Securities Regulatory Commission (CSRC) and stock exchanges require firms to issue a performance forecast if they report a net loss, expect a profit or loss change of 50% or more relative to the previous year, turn profitable, or end the fiscal period with negative net assets.

¹⁹ In China, regulatory guidance on the content and structure of management forecasts is relatively flexible. Firms are encouraged to discuss expected performance ranges, underlying drivers, and sources of uncertainty, but retain substantial discretion in how these elements are presented. As a result, management forecasts exhibit considerable cross-sectional variation in both length and topical coverage, which makes them well suited for examining heterogeneity in disclosure informativeness.

TABLE 3
Firm-Specific Queries around Corporate Information Events

Dep. Var. =	(1)	(2)	(3)	(4)
	# <i>Specific_Queries</i>			
Company disclosures:				
<i>EA_Pos</i>	0.070** (2.43)	0.014 (0.53)	0.014 (0.51)	0.014 (0.52)
<i>EA_Neg</i>	0.131*** (4.17)	0.060 (1.56)	0.060 (1.55)	0.060 (1.56)
<i>MF_Pos</i>	0.140 (1.18)	0.086 (0.77)	0.284*** (2.97)	0.254** (2.40)
<i>MF_Neg</i>	0.285 (1.52)	0.236 (1.27)	0.350 (1.44)	0.186 (1.35)
<i>Event_Other</i>	0.068*** (4.18)	0.046*** (3.33)	0.046*** (3.32)	0.046*** (3.32)
Information intermediaries:				
<i>AnaReport</i>		0.020 (0.74)	0.021 (0.76)	0.020 (0.75)
<i>Media</i>		0.149*** (8.67)	0.149*** (8.63)	0.149*** (8.64)
Cross-sectional effects: Disclosure informativeness				
<i>MF_Pos</i> × <i>More_Topics</i>			−0.456** (−2.27)	
<i>MF_Neg</i> × <i>More_Topics</i>			−0.351 (−1.39)	
<i>MF_Pos</i> × <i>Longer</i>				−0.396* (−1.87)
<i>MF_Neg</i> × <i>Longer</i>				0.103 (0.49)
Observations	596,405	596,405	596,405	596,405
Pseudo <i>R</i> ²	0.668	0.669	0.669	0.669
Firm FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes

This table reports OLS regression results on the association between firm-specific queries and corporate information events, using firm-day observations from the first half of 2024. The dependent variable, *#Specific_Queries*, measures the number of firm-specific queries. Columns (1) and (2) report the baseline results, whereby information events include either corporate disclosures only in Model (1) or both corporate disclosures and information releases by intermediaries in Model (2). For corporate disclosures, we include separate daily indicator variables for five types of events: positive earnings announcements (*EA_Pos*), negative earnings announcements (*EA_Neg*), positive management forecasts (*MF_Pos*), negative management forecasts (*MF_Neg*), and other (unsigned) material events, such as mergers and acquisitions (*Event_Other*). We additionally identify firm-days with intermediaries' information releases using indicator variables for analyst report releases (*AnaReport*) and for media coverage (*Media*). In Columns (3) and (4), the management forecast indicators are interacted with variables capturing two cross-sectional characteristics of the forecasts. Specifically, *More_Topics* indicates forecasts in which a larger number of topics are discussed than in the sample median forecast, and *Longer* indicates forecasts with a word count above the sample median forecast. All variables are defined in Appendix A. *t*-statistics in parentheses are based on standard errors clustered by firm and date. *, **, and *** indicate statistical significance at the 0.10, 0.05, and 0.01 levels, respectively.

more credible (Skinner, 1994), thereby diminishing investors' incentives to engage in further information searches. We find no analogous moderation patterns for negative management forecasts.

Additional analyses indicate that firm-related information events coincide with systematic shifts in the content of GenAI queries, including both the topics users focus on and the tasks they perform. For example, around

earnings announcements, queries often center on financial performance, whereas periods with negative management forecasts are associated with increased attention to contextual, nonfinancial aspects such as business outlook, operations, and competition. These results, together with further task-level evidence, are reported in online appendix OA.11.

4. User Interactions with GenAI

In the second part of our analysis, we examine how users interact with GenAI tools and how these interactions reflect perceived usefulness. We focus on two dimensions of user responses: (1) immediate feedback and subsequent engagement associated with different attributes of GenAI answers and (2) evidence of user learning from early interactions with the platform.

4.1 CONSTRUCTION OF ANSWER ATTRIBUTES

We construct a parsimonious set of answer-level attributes to capture the informational content, tone, and complexity of GenAI responses; details of this process are provided in online appendix OA.12. Jointly, these measures are selected to reflect whether answers address users' informational needs in ways that are likely associated with perceived usefulness and subsequent engagement.

First, we construct a measure of signal alignment (denoted *Signal Align*), which captures whether the directional trading signal embedded in an answer is consistent with the firm's subsequent short-term stock performance. Using a commercial LLM, we classify each answer's trading signal as positive, neutral, or negative. We then rank firms within each industry based on their future seven-day returns and define an indicator variable equal to one if the signal aligns with realized performance.²⁰

Second, we construct language-based attributes that capture forward-looking content, subjectivity, and sentiment, using a fine-tuned Chinese BERT model following Bozanic et al. (2018) and established sentiment classification approaches (Merkley, 2014; Huang et al., 2014). All language measures are scaled by the total number of sentences in the answer.

Finally, we proxy for informativeness and complexity using readability (Gunning-Fog index), topic diversity, specificity, reference intensity, and answer length. Topic diversity measures the extent to which an answer covers topics not explicitly requested in the query, while specificity, references, and word count capture increasing levels of content granularity and information density.

²⁰ This comparison relies on relative return rankings within the same industry over a seven-day horizon, motivated by the high turnover rate in Chinese stock markets (Allen et al., 2024).

4.2 ANSWER ATTRIBUTES, USER FEEDBACK, AND CONTINUED ENGAGEMENT

We next examine how answer attributes relate to users' immediate reactions and subsequent engagement with the GenAI platform. We focus on positive and negative user reactions, as well as continued usage, measured by the number of future queries submitted after the current answer. Positive reactions include copying, thumbs-up, and sharing, while thumbs-down proxies for a negative reaction.

We begin by examining the correlations between GenAI answer attributes and query attributes. To this end, we compute the three language attributes, *Specificity*, and raw word count for both queries and answers, and report their correlations in table 4, panel A. For each language attribute, the corresponding attribute in the query is positively correlated with that in the answer, suggesting that GenAI responses partially reflect users' language choices. In contrast, query length is negatively correlated with both answer length and answer specificity. Together with the negative correlation between query specificity and answer length of -0.099 , these patterns indicate that short and unspecific queries tend to elicit broader and more expansive answers.

We proceed to analyze which answer attributes users perceive as more useful by estimating the following model:

$$\begin{aligned} Feedback_a = & \beta_1 Signal_Align_a + \beta_2 Forward_Looking_a + \beta_3 Subjectivity_a \\ & + \beta_4 Sentiment_a + \beta_5 Fog_a + \beta_6 Topic_Diversity_a \\ & + \beta_7 Specificity_a + \beta_8 \log(1 + \#References_a) \\ & + \beta_9 \log(1 + \#Words_a) + Date\ FE_t + \epsilon_a, \end{aligned} \quad (2)$$

where the dependent variable captures either users' immediate reactions to the GenAI answer or their subsequent engagement with the platform. Immediate reactions include positive feedback (copying, thumbs-up, or sharing) and negative feedback (thumbs-down). Subsequent engagement is the number of stock-market-related queries submitted by the same user after the current answer, denoted *#Future_Queries*. The immediate feedback measures reflect timely, answer-level reactions by a single user, whereas *#Future_Queries* serves as a proxy for longer horizon perceived usefulness. We include date fixed effects to absorb common time-varying shifts in user sentiment or platform usage. When *#Future_Queries* is the dependent variable, we estimate the model using a Poisson specification.

Table 4, panel B, reports regression results at the individual-answer level. Signal alignment is positively associated with immediate positive feedback, indicating that users are able to assess the usefulness of directional information shortly after reading an answer. We find no corresponding increase in thumbs-down reactions, suggesting that misaligned signals are not systematically penalized with negative feedback.

TABLE 4
Answer Attributes, User Feedback, and Continued Engagement

Panel A: Correlations of query attributes (in columns) with answer attributes (in rows)			
Answer	Query		
	Forward_Looking	Subjectivity	Sentiment
Forward_Looking	0.383***	0.235***	0.046***
Subjectivity	0.234***	0.420***	0.070***
Sentiment	0.171***	0.110***	0.343***
Specificity	-0.159***	-0.187***	-0.021***
Log(1+#Words)	0.060***	0.108***	-0.023***
Panel B: Regression results on user feedback and engagement			
Dep. Var. =	(1)	(2)	(3)
	Positive_FB	Thumbs_Down	#Future_Queries
Alignment between trading signals and short-term stock returns:			
Signal_Align (Future 7 days)	0.296* (1.74)	-0.001 (-0.12)	0.124*** (6.27)
Language attributes:			
Forward_Looking	-1.130 (-1.54)	0.002 (0.09)	0.360** (2.35)
Subjectivity	6.879*** (11.74)	0.031 (1.14)	0.157 (0.94)
Sentiment	1.951*** (3.95)	-0.039*** (-2.88)	0.291*** (3.39)
(Continued)			

TABLE 4—(Continued)

Panel B: Regression results on user feedback and engagement			
Dep. Var. =	(1) <i>Positive_FB</i>	(2) <i>Thumbs_Down</i>	(3) <i>#Future_Queries</i>
Informativeness and complexity:			
<i>Fog</i>	0.111** (2.50)	0.002 (1.08)	−0.002 (−0.18)
<i>Topic_Diversity</i>	−9.301*** (−2.64)	−0.145*** (−2.73)	−1.815 (−1.54)
<i>Specificity</i>	−4.035*** (−2.97)	0.094* (1.68)	−0.370 (−1.11)
<i>Log(1+ #References)</i>	−7.323*** (−14.78)	−0.028*** (−4.40)	−0.360** (−1.98)
<i>Log(1+ #Words)</i>	5.268*** (11.94)	0.007 (0.76)	−0.138* (−1.91)
Observations	376,491	376,491	328,676
(Pseudo) <i>R</i> ²	0.071	0.001	0.387
Date FE	YES	YES	YES

This table reports (1) correlations between answer attributes (rows) and query attributes (columns) in panel A and (2) regression estimates on the association between answer attributes and user feedback or continued engagement in panel B. The analysis in panel B is conducted at the answer level. The first two dependent variables are indicator variables for whether the current answer receives positive feedback (including thumbs-up, sharing via a link, or copying) or negative feedback (i.e., thumbs-down). Column 3 focuses on users who continue to engage with the platform by submitting more than one stock-market-related query and excludes their initial queries from the estimation sample. The dependent variable *#Future_Queries* denotes the number of subsequent queries following the current answer. All variables are defined in appendix A. *t*-statistics are shown in parentheses, with standard errors clustered by user. *, **, and *** indicate statistical significance at the 0.10, 0.05, and 0.01 levels, respectively.

Several language attributes are also related to user reactions. Answers containing more subjective language and a more positive tone are more likely to receive positive feedback, consistent with users placing value on qualitative or interpretive content. We further examine continued engagement among users who return to the platform after their initial interaction. For these continuing users, signal alignment is strongly positively associated with future query volume, indicating a higher likelihood of re-engagement when prior answers appear directionally informative. Forward-looking content and subjectivity are also positively related to continued usage, whereas measures of complexity are negatively related, consistent with users' aversion to overly dense or difficult-to-process content (Umar, 2022).²¹

4.3 USER LEARNING FROM EARLY INTERACTIONS

We examine whether and how users adapt their subsequent queries in response to information obtained from earlier interactions with GenAI. We capture this potential learning behavior by focusing on two features of later queries: their specificity and their use of financial metrics. These measures serve as indicators of whether users expand their technical knowledge, factual understanding, or financial vocabulary following exposure to more informative GenAI answers. We hypothesize that users are more likely to incorporate (i) specific entities (such as people, locations, or organizations) and (ii) financial metrics related to technical financial statement analysis in later queries when earlier answers contain richer information along these dimensions. To test this hypothesis, we estimate the following equation:

$$\begin{aligned}
 \text{Attribute_Post3_Queries}_n = & \beta_1 \text{Attribute_First3_Answers}_n \\
 & + \beta_2 \text{Attribute_First3_Queries}_n \\
 & + \beta_3 \text{Sentiment_Post3_Queries}_n \\
 & + \beta_4 \text{Subjectivity_Post3_Queries}_n \\
 & + \beta_5 \text{Forward_Post3_Queries}_n \\
 & + \beta_6 \log(1 + \#Words_Post3_Queries_n) \\
 & + \beta_7 \log(1 + \#Queries_Post3_n) + \epsilon_n. \quad (3)
 \end{aligned}$$

The analysis is conducted at the individual-user (n) level and examines how attributes of a user's later queries (specificity and financial metrics usage) relate to these attributes of the GenAI's answers to that same user's earlier queries. We control for early query attributes to account

²¹ We exclude first-time usage in analyses of continued engagement. Results for first-time users are qualitatively different and indicate weaker sensitivity to answer attributes. To address concerns that immediate follow-up queries may reflect dissatisfaction with an answer rather than sustained engagement, we impose alternative timing restrictions on future queries by excluding immediate follow-ups. The results remain qualitatively similar across these alternative definitions. Details on these robustness checks are in online appendix OA.13.

for users' initial style and prior knowledge. The following dependent variables are measured using queries submitted from the fourth query onward: (i) the average specificity (*Specificity_Post3_Queries*); (ii) the average probability that a query mentions *any* financial metric (*Any_FM_Post3_Queries*); and (iii) three separate indicator variables (*Specific_FM_Post3_Queries*) for mentioning a specific financial metric: ROE, solvency, or accounts receivable.²²

The main independent variables are the same attributes measured over the first three queries and, separately, in the corresponding GenAI answers. For example, *Specific_FM_First3_Queries* and *Specific_FM_First3_Answers* capture the average incidence of a specific financial metric in the first three queries and in the answers to those queries, respectively. We further control for all three linguistic attributes of later queries, including sentiment, subjectivity, and forward-looking orientation. To mitigate truncation bias arising from the fixed end date of our sample (end of June), we include fixed effects for the month of the user's entry into the platform.

The results in table 5 indicate that early interactions with GenAI are meaningfully associated with the attributes of later queries. When the first three GenAI answers contain more financial metrics or are of higher specificity, subsequent queries submitted by the same user are more likely to mention financial metrics and to be more specific. These associations hold after controlling for the same attributes of the first three queries; that is, they are incremental to the expectedly high persistence in a user's language style and her financial metric usage. Moreover, when the first three answers referenced a specific financial metric (ROE, solvency, or accounts receivable), users are more likely to mention that very metric in a later query.

We acknowledge that users may have been familiar with these metrics prior to interacting with the platform. However, because we condition on users' early query patterns, the incremental associations suggest that GenAI may direct users' attention toward such information in subsequent searches. We also emphasize that these analyses cannot establish whether the observed patterns translate into learning benefits, such as improved answer quality or superior trading outcomes. Rather, we can only infer that users' continued incorporation of these elements in later queries reflects a *perceived* benefit. Overall, the evidence indicates that users' subsequent query behavior is systematically related to the content of GenAI answers encountered during their early interactions with the platform.

²² We randomly select these three example financial metrics from the list of the most frequently used financial metrics reported in online appendix OA.14.

TABLE 5
User Learning from Early Interactions with GenAI

Dep. Var. =	(1)	(2)	(3)	(4)	(5)
Specific Financial Metric =	Specificity_Post3_Queries	Any_FM_Post3_Queries		Specific_FM_Post3_Queries	
Specificity_First3_Answers	0.081*** (7.41)				
Specificity_First3_Queries	0.352*** (14.06)				
Any_FM_First3_Answers		0.055*** (7.54)			
Any_FM_First3_Queries		0.392*** (13.78)			
Specific_FM_First3_Answers			0.051*** (5.59)	0.078*** (7.27)	0.073*** (4.97)
Specific_FM_First3_Queries			0.322*** (22.24)	0.422*** (15.91)	0.414*** (12.38)
Sentiment_Post3_Queries	-0.014** (-3.59)	-0.028*** (-13.57)	-0.010* (-2.52)	-0.016*** (-4.27)	-0.031*** (-5.15)

(Continued)

TABLE 5—(Continued)

Dep. Var. =	(1)	(2)	(3)	(4)	(5)
	<i>Specificity_Post3_Queries</i>	<i>Any_FM_Post3_Queries</i>		<i>Specific_FM_Post3_Queries</i>	
Specific Financial Metric =			ROE	Solvency	Accounts Receivable
<i>Subjectivity_Post3_Queries</i>	−0.060*** (−22.81)	−0.055*** (−4.77)	0.000 (0.03)	0.016*** (6.01)	0.000 (0.05)
<i>Forward_Post3_Queries</i>	−0.052*** (−20.35)	−0.058*** (−12.51)	−0.025*** (−16.51)	−0.020*** (−11.79)	−0.014*** (−9.56)
<i>Log(1+Words_Post3_Queries)</i>	0.006** (2.75)	0.023*** (5.92)	0.039*** (12.02)	0.034*** (8.78)	0.030*** (17.37)
<i>Log(1+#Queries_Post3)</i>	−0.028*** (−14.37)	0.093*** (28.62)	0.026*** (9.47)	0.018*** (6.14)	0.026*** (10.57)
Observations	78,414	78,887	78,887	78,887	78,887
R ²	0.227	0.371	0.119	0.130	0.155
Entry-Month FE	Yes	Yes	Yes	Yes	Yes

This table examines the association between selected attributes of early GenAI answers, defined as the answers to a user's first three queries (*First_3*), and the characteristics of that user's subsequent queries from the fourth query onward (*Post_3*). The dependent variables are constructed at the user level using all *Post_3* queries. Specifically, we measure: (i) *Specificity_Post3_Queries*, defined as the average query specificity across *Post_3* queries, where query specificity is measured as the number of specific words scaled by total words; (ii) *Any_FM_Post3_Queries*, defined as the average share (proportion) of *Post_3* queries that mention at least one financial metric; and (iii) *Specific_FM_Post3_Queries*, a placeholder for three separate dependent variables, each equal to one if a specific financial metric, namely "ROE," "Solvency," or "Accounts Receivable," appears at least once in *Post_3* queries, and zero otherwise. These three financial terms are randomly drawn from the set of the most frequently appearing financial metrics reported in online appendix OA.14. All dependent variables are measured using all queries submitted by a given user after the first three (i.e., beginning with the fourth query). The independent variables include the same set of attributes measured across the first three user queries and, separately, across the corresponding first three GenAI answers to those three early queries. For instance, *Specific_FM_First3_Queries* (*Specific_FM_First3_Answers*) is an indicator equal to one if the focal financial metric (ROE, Solvency, or Accounts Receivable) appears at least once in the user's first three queries (in the answers to the first three queries), and zero otherwise. All regression specifications additionally control for average linguistic attributes of the user's subsequent queries. All variables are defined in appendix A. *t*-statistics are shown in parentheses and are based on standard errors clustered by the entry month. *, **, and *** indicate statistical significance at the 0.10, 0.05, and 0.01 levels, respectively.

5. Preliminary Associations Between Queries and Market Outcomes

5.1 USER QUERIES AND STOCK MARKET TRADING

In the third part of our analysis, we examine preliminary associations between user query activity and market trading outcomes. We further explore cross-sectional variation in these associations as a function of users' perceived usefulness of GenAI answers, as reflected in their immediate feedback.

As we cannot observe the individual stock trades following or based on user engagement with GenAI, we examine the association between *aggregate* platform activity and *aggregate* (market-wide) trading outcomes using the following regression model:

$$\begin{aligned} \text{Trading Outcomes}_{i,t} = & \beta \log(1 + \# \text{Specific_Queries})_{i,t} + \psi \log(MV)_{i,t} \\ & + \alpha \text{Company Disclosures}_{i,t} \\ & + \gamma \text{Information Intermediaries}_{i,t} \\ & + \delta \text{Other Information Channels}_{i,t} \\ & + \text{Firm FE}_i + \text{Date FE}_t + \epsilon_{i,t}. \end{aligned} \quad (4)$$

The dependent variables represent different trading outcomes: the daily bid-ask spread, the Amihud illiquidity measure, abnormal trading volume, and the volume-synchronized probability of informed trading (*VPIN*). The daily abnormal bid-ask spread (*AbnSpread*) is calculated as the value-weighted average of the effective bid-ask spread, scaled by the mid-quote (Holden and Jacobsen, 2014), minus the average scaled spread over the preceding 20 trading days $[-20, -1]$. Higher abnormal spreads are generally interpreted as indicative of greater price protection against adverse selection and higher information asymmetry (Glosten and Milgrom, 1985; Welker, 1995). The Amihud (2002) illiquidity measure is computed as the daily ratio of absolute stock return to trading volume. Abnormal trading volume (*AbnVol*) is defined as the daily stock turnover rate minus its average over the preceding 20 trading days $[-20, -1]$, capturing market participants' attention and trading activity in response to new information (Barber and Odean, 2008). Finally, *VPIN* is calculated according to Easley et al. (2012) and serves as a proxy for adverse selection risk and the presence of informed trading in the market.

We control for firms' (daily) market value of equity (*MV*; in log form) and include indicator variables for all previously examined firm disclosure events (separately for positive and negative news) as well as the number of analyst reports and news releases in the media. Additionally, we control for potentially confounding investor information activities on search engines (Drake et al., 2012) and questions posted on public investor platforms (Lee and Zhong, 2022).

TABLE 6
User Queries and Market Trading

Dep. Var. =	(1) <i>AbnSpread</i>	(2) <i>Amihud Illiquidity</i>	(3) <i>AbnVol</i>	(4) <i>VPIN</i>
<i>Log(1+#Specific_Queries)</i>	0.296*** (10.35)	0.235*** (4.42)	0.325*** (18.93)	0.103*** (4.42)
<i>Log(MV)</i>	-3.763*** (-9.71)	-8.768*** (-14.63)	0.982*** (4.93)	-1.225*** (-3.96)
Company disclosures:				
<i>EA_Pos</i>	0.519*** (6.29)	0.357** (2.53)	0.494*** (7.18)	1.101*** (9.88)
<i>EA_Neg</i>	0.543*** (5.35)	0.143 (0.67)	0.217*** (4.85)	1.216*** (9.01)
<i>MF_Pos</i>	0.223** (2.15)	-0.658 (-1.30)	0.296* (1.66)	1.148*** (6.57)
<i>MF_Neg</i>	0.925*** (6.10)	1.021*** (6.03)	-0.151*** (-2.63)	0.069 (0.59)
<i>Event_Other</i>	0.092*** (4.55)	0.054* (1.79)	0.111*** (8.47)	-0.018 (-0.72)
Information intermediaries:				
<i>AnaReport</i>	0.026 (0.34)	-0.142 (-1.47)	0.124*** (4.96)	0.137** (2.28)
<i>Media</i>	0.280*** (13.64)	0.177*** (6.86)	0.313*** (16.60)	0.227*** (8.56)
Other information channels:				
<i>Log(1+SVI)</i>	0.263*** (3.91)	-0.119 (-1.35)	1.178*** (8.86)	0.340*** (4.93)
<i>Log(1+Retail Platform)</i>	0.151*** (6.97)	-0.003 (-0.11)	0.290*** (15.11)	0.096*** (3.65)
Observations	594,699	596,989	595,553	596,989
<i>R</i> ²	0.285	0.565	0.124	0.141
Firm FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes

This table presents regression results on the association between user queries and market trading outcomes. The dependent variables are abnormal bid-ask spread (*AbnSpread*), Amihud illiquidity, abnormal trading volume (*AbnVol*), and *VPIN*. *AbnSpread* is computed as the value-weighted average of the effective bid-ask spread, scaled by the mid-quote (Holden and Jacobsen, 2014), minus the average of the scaled spread over trading days [-20,-1]. The Amihud illiquidity measure is calculated as the daily ratio of the absolute stock return to its trading volume (Amihud, 2002). *AbnVol* is defined as the daily stock turnover rate minus its average over trading days [-20,-1]. *VPIN* captures the volume-synchronized probability of informed trading, following Easley et al. (2012). The key explanatory variable is *Log(1+#Specific_Queries)*. We include the same indicators for information events as in table 3, covering both corporate disclosure events and information releases by intermediaries; we additionally control for (the logarithm of) the search volume index and the number of questions posted on public investor Q&A platforms maintained by stock exchanges (Lee and Zhong, 2022). All variables are defined in appendix A. *t*-statistics are shown in parentheses, with standard errors clustered by firm and date. *, **, and *** indicate statistical significance at the 0.10, 0.05, and 0.01 levels, respectively.

The results of estimating equation (4) in table 6 reveal a positive association between user queries and both abnormal bid-ask spreads and the Amihud illiquidity measure (columns 1 and 2). Taken together, these results indicate a positive association between platform query activity and lower stock liquidity. We note that these results are consistent with two possible explanations. First, GenAI usage may help noise traders detect their in-

formational disadvantage and thus deter them from trading (Peress and Schmidt, 2020). A second possible explanation is that GenAI usage enables some, but not all, investors to become more informed, who then trade more aggressively and thus widen bid-ask spreads and reduce liquidity.²³

We try to empirically disentangle these two explanations by examining the abnormal trading volume (column 3) and the (volume-synchronized) probability of informed trading as proxied by *VPIN* (column 4). The positive association between user queries and abnormal trading volume is inconsistent with the first explanation of reduced noise trading. Conversely, we find a positive association between user queries and *VPIN*, which is consistent with an interpretation in which higher GenAI activity coincides with increased informed trading by a subset of investors. As such, AI activity appears to be associated with greater information asymmetry among investors, that is, with widening the gap between informed and uninformed traders (Glosten and Milgrom, 1985).

We caution that these empirical associations may be impacted by a selection bias insofar as users who find GenAI particularly beneficial are also more likely to engage with the platform, further amplifying the information advantage of platform users relative to nonusers. Additionally, within the platform's user base, more sophisticated traders may possess better technological literacy and incur lower learning costs for both processing information and navigating the platform. For example, such investors may have better strategies for prompting, searching, and integrating information, which complements their prior (already superior) knowledge.

Our findings join the ongoing discussion of how GenAI impacts market participants' information processing and trading activities. Two concurrent studies by Cheng et al. (2025) and Bertomeu et al. (2025) exploit ChatGPT outages and the ban of ChatGPT in Italy as exogenous events, respectively, and document mixed effects on market liquidity. Cheng et al. (2025) find that GenAI outages are associated with a decrease in bid-ask spreads, aligned with our associations. In contrast, Bertomeu et al. (2025) document that bid-ask spreads widen during the Italian ban, suggesting a deterioration in liquidity. Although our design does not permit causal inferences, the results offer insights from a novel perspective by directly observing cross-day and cross-firm variation in user activity on a GenAI platform. We hope this study advances our understanding of how GenAI shapes the information environment of market participants and encourages future research on potential disparities in market trading associated with GenAI usage.

²³ In principle, it is also possible that GenAI usage might lead to an increase in noise trading that indirectly *consumes* liquidity through imposing higher inventory risks on market makers, specifically risks associated with future systematic position unwinding (Grossman and Miller, 1988; Foucault et al., 2011). We note, however, that this potential explanation, adopted from the disclosure setting in Friedman et al. (2025), is less likely to hold in our GenAI setting, as the platform users are largely retail investors who will not be able or willing to coordinate trading activities among themselves.

5.2 TRADING OUTCOMES CONDITIONAL ON ANSWER FEEDBACK AND SENTIMENT

This section examines the extent to which user reactions to GenAI answers moderate the association between query volume and market trading outcomes. We begin by computing the proportion of answers with positive user feedback (including thumbs-up, shares, or copies) out of the total number of answers on a given day, denoted *PosFBRatio*. We then interact this variable with the daily query volume and include this interaction term in equation (4) above.

Panel A of table 7 reports the results.²⁴ Across all four specifications, the coefficients on the interaction terms between query volume and the positive feedback ratio are positive, albeit statistically insignificant in the regression for abnormal trading volume. The estimates of the other three specifications point to stronger associations between query activity and market illiquidity as well as indicators of informed trading when a larger share of answers receives positive user feedback.

These patterns suggest that periods in which GenAI answers are perceived more favorably by users tend to coincide with larger associations between platform activity and trading outcomes. One possible interpretation is that users are more likely to rely on, or further engage with, information obtained from answers they regard as useful. At the same time, we emphasize that these results are purely associational. The observed interaction effects may also reflect unobserved factors that jointly influence user feedback, query activity, and market conditions, and thus should not be interpreted as evidence of a causal effect.

Next, we examine whether the sentiment embedded in GenAI answers is associated with (signed) daily abnormal returns using the following regression model:

$$\begin{aligned} AbnReturn_{i,t} = & \beta Avg_Sentiment_{i,t} + \alpha CompanyDisclosures_{i,t} \\ & + \gamma InformationIntermediaries_{i,t} \\ & + \delta OtherInformationChannels_{i,t} \\ & + Firm\ FE_i + Date\ FE_t + \epsilon_{i,t}. \end{aligned} \quad (5)$$

The dependent variable is the daily abnormal return, calculated as the difference between the actual return and the benchmark return implied from a firm-specific time-series regression of daily stock returns on daily market returns over the $[-31, -1]$ trading-day window, following Ke et al. (2003). *Avg_Sentiment* denotes the firm-day average sentiment across all GenAI answers. We further decompose this measure into sentiment for answers that receive positive user feedback (*Avg_Sentiment_PosFB*) and those that do not (*Avg_Sentiment_NonPosFB*).

²⁴ The sample size in this analysis is smaller than that in table 6 because these tests condition on the availability of at least one user query, as the feedback ratios are undefined in those cases.

TABLE 7
Positive Answer Feedback and Market Outcomes

Panel A: User queries and market trading conditional on positive answer feedback				
Dep. Var. =	(1) <i>AbnSpread</i>	(2) <i>Amihud Illiquidity</i>	(3) <i>AbnVol</i>	(4) <i>VPIN</i>
<i>Log(1+#Specific_Queries)</i>	0.192*** (8.39)	0.151*** (4.99)	0.233*** (10.18)	0.078** (2.59)
<i>Log(1+#Specific_Queries)</i> $\times$ <i>PosFBRatio</i>	0.072** (1.99)	0.070** (2.03)	0.028 (0.78)	0.191*** (2.71)
Observations	149,148	150,309	150,177	150,309
R^2	0.273	0.571	0.201	0.151
<i>Log(MV)</i>	Yes	Yes	Yes	Yes
Company Disclosures	Yes	Yes	Yes	Yes
Information Intermediaries	Yes	Yes	Yes	Yes
Other Information Channels	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Date FE	Yes	Yes	Yes	Yes
Panel B: Answer sentiment and abnormal returns				
Dep. Var. =	(1)		(2)	
	<i>AbnReturn</i>			
<i>Avg_Sentiment</i>	0.601*** (12.91)			
<i>Avg_Sentiment_PosFB</i> (β_1)			0.809*** (9.36)	
<i>Avg_Sentiment_NonPosFB</i> (β_2)			0.438*** (4.48)	
p -value for $\beta_1 = \beta_2$			<0.01	
Observations	142,211		22,714	
R^2	0.092		0.213	
<i>Log(MV)</i>	Yes		Yes	
Company Disclosures	Yes		Yes	
Information Intermediaries	Yes		Yes	
Other Information Channels	Yes		Yes	
Firm FE	Yes		Yes	
Date FE	Yes		Yes	

This table reports regression results on the association between user queries and market trading outcomes, conditional on the extent of positive user feedback on the platform's answers. In panel A, *PosFBRatio* is defined as the proportion of answers that receive positive feedback—including thumbs-up, shares, or copies—out of all answers provided for a given firm on a given day. In panel B, the dependent variable is the daily abnormal return, calculated as the residual from firm-specific regressions of daily stock returns on daily market returns estimated over the $[-31, -1]$ window (Ke et al., 2003). The key independent variable is the daily average sentiment of all responses to user queries (*Avg_Sentiment*). We additionally compute average sentiment separately for answers that receive positive feedback and those that do not. Specifically, *Avg_Sentiment_PosFB* and *Avg_Sentiment_NonPosFB* denote the average sentiment of answers with and without positive feedback, respectively. All variables are defined in appendix A. t -statistics are shown in parentheses, with standard errors clustered by firm and date. *, **, and *** indicate statistical significance at the 0.10, 0.05, and 0.01 levels, respectively.

Table 7, panel B, reports the results from estimating equation (5). We document a significantly positive association between the sentiment conveyed in GenAI answers and same-day abnormal stock returns. Column 1 shows that *Avg_Sentiment* is positively correlated with abnormal returns, indicating that more favorable sentiment in GenAI answers tends to coincide with higher contemporaneous stock performance. Column 2 further shows that this association is stronger when the answers receive positive user feedback, with the difference significant at the 1% level.

These results are consistent with a setting in which users place greater weight on sentiment signals embedded in answers they perceive as useful. An alternative interpretation is that GenAI answers reflect prevailing market sentiment in real time, such that the observed sentiment-return relation arises because both respond to the same underlying information environment. Under this interpretation, GenAI acts as a conduit that aggregates and echoes existing market sentiment rather than exerting an independent influence on trading behavior.

Overall, the evidence indicates that sentiment embedded in GenAI answers is systematically related to daily stock returns. However, as with our other market-level analyses, these correlations do not permit causal interpretation and should be viewed as descriptive patterns that motivate future research with tighter identification.

6. Pathways for Future Research

This section builds on our three sets of empirical findings—heterogeneity in query behavior, user interactions with the GenAI platform, and preliminary associations between query activity and stock market outcomes—to outline promising directions for future research.

First, our descriptive evidence reveals substantial heterogeneity in query topics and tasks across user subgroups with differing levels of AI familiarity and financial literacy. Future research could extend this analysis along several avenues. One avenue is to examine the determinants of these differences, including variation in firms' and users' information environments, such as firm complexity, news intensity, and uncertainty, as well as investors' opportunity sets, including their time constraints and access to competing information sources. A second avenue relates to timing. While our evidence documents systematic variation in usage around corporate information events, it remains unclear how the nature of tasks evolves around such events: Ex ante, pre-announcement tasks presumably focus on information gathering, whereas post-announcement tasks may be more about interpretation and verification. Relatedly, the relatively limited use of summarization tasks, despite their natural fit with information overload and the demonstrated ability of LLMs to distill long-form disclosures, points to potentially testable frictions in how users deploy GenAI. Third, our analysis focuses on heterogeneity within users of a single platform and, by construction, largely within retail investors. The broader impact of GenAI across

other market participants, such as institutional investors or financial analysts remains an open question.

Second, our evidence on user-GenAI interactions highlights both systematic user preferences and potential learning dynamics. Future research could more explicitly model interaction sequences and user utility, for example, by examining repeated prompting, follow-up questions, verification behavior, and ending the search as outcomes of users' perceived costs and benefits from continued interaction. An important dimension of these interactions pertains to the accuracy of GenAI outputs, including whether hallucinations, errors, or systematic biases affect subsequent user behaviors, and whether exposure to more accurate or more balanced answers shapes future interaction patterns. From a platform design perspective, further work could investigate whether features such as feedback mechanisms, adaptive content presentation, or structured templates for common investor tasks facilitate more effective and efficient use of GenAI. Finally, although our results indicate that users' later queries increasingly incorporate financial terminology appearing in earlier GenAI answers, we cannot assess whether this adaptation reflects meaningful learning that translates into economic benefits for users, such as improved trading performance.

Third, our exploratory analyses indicate that simulated stock market returns prior to earnings announcements vary by query type, with queries involving financial statement analysis and finance-related topics associated with higher simulated returns (online appendix table OA.8). These associations do not, however, distinguish between alternative mechanisms. Higher returns could reflect a comparative advantage of LLMs in processing structured financial information, but they could also arise from user selection, whereby more financially sophisticated investors both submit such queries and achieve higher returns irrespective of GenAI usage. Future research aimed at disentangling treatment effects from selection effects would help clarify how GenAI can enhance the effective participation of less sophisticated investors.

Taking a broader perspective, our findings also motivate extensions that view GenAI as a general-purpose technology reshaping the information environment. Our setting captures an early stage of GenAI development and GenAI adoption, and the results cannot speak to an equilibrium of more pervasive and sustained use. For example, GenAI may either reinforce or erode the informational advantages of sophisticated investors, raising important distributional and welfare considerations for investors and regulators alike. On the information supply side, firms and intermediaries adopting GenAI likely adjust their disclosure practices as the costs of collecting, synthesizing, and disseminating information decline. This line of reasoning suggests that supplier responses potentially mitigate efficiency gains on the demand side. However, it is also possible that firms develop more structured and machine-readable disclosure formats that better accommodate AI-assisted information processing by the recipients of the information, potentially amplifying the benefits of their GenAI use.

Finally, we call for more rigorous empirical identification strategies that allow for causal inference on the effects of GenAI usage, in particular, on individual trading behaviors and market-level outcomes. Future studies could exploit plausibly exogenous variation, including regulatory changes, staggered rollouts of GenAI tools, or randomized field and laboratory experiments to assess the implications of GenAI for investor decision-making and price efficiency.

7. Conclusion

This study provides early descriptive evidence on how retail investors use GenAI for stock-related information processing. Drawing on query-level data from one of China's largest GenAI platforms, we document how users engage with this emerging technology across topics, tasks, and time, and how these patterns vary across individuals and information environments. We further examine how attributes of GenAI answers relate to user feedback and continued engagement, and how aggregate platform activity correlates with market-level trading variables.

Our analyses show that users primarily submit firm-specific queries that emphasize information integration, particularly around business outlook, financial performance, and production and operations. Platform usage, however, is uneven. Engagement varies substantially across regions and over time, and user retention is generally low, suggesting that early adoption is characterized by episodic trials rather than sustained use. More engaged and financially sophisticated users are disproportionately active in analytical and integrative tasks, while less engaged users tend to rely on GenAI for broader information awareness. Query activity also increases around major corporate information events, although much of this variation parallels contemporaneous media coverage rather than firm disclosures alone. We further document evidence consistent with a substitutive relationship between detailed management forecasts and same-day GenAI usage.

We also find that attributes of GenAI answers are closely related to user feedback and continued engagement. Answers whose directional signals align with subsequent stock performance receive more positive immediate feedback and are associated with greater subsequent usage, particularly among continuing users. We view these results as indicative of both short-term and long-term perceived usefulness of the platform. At the same time, users appear to prefer concise and accessible responses over longer or more complex answers. Tracking users over time, we find that attributes of early GenAI answers are reflected in later queries, consistent with users adjusting their information focus and terminology following initial interactions with the platform.

Finally, we explore how aggregate GenAI usage relates to market-level trading variables. Higher query activity is associated with wider bid-ask spreads, higher abnormal trading volume, and greater measures of informed trading, suggesting increased information heterogeneity across

investors. In addition, sentiment embedded in GenAI answers is positively associated with same-day abnormal returns, particularly when those answers receive positive user feedback. Although these findings are associational in nature and do not permit causal inference, they highlight how GenAI activity is intertwined with information flows and trading behavior in financial markets.

Taken together, our results shed light on how GenAI is beginning to reshape information processing among retail investors. They point to important heterogeneity in users of these tools, how they are used, and the contexts in which they appear most relevant. For platform providers and regulators, understanding these patterns may help inform design choices and educational initiatives aimed at improving effective use. Regarding firms, our findings suggest that the interaction between corporate disclosure and AI-assisted information processing may become an increasingly relevant consideration in shaping investor communication strategies.

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APPENDIX A: VARIABLE DEFINITIONS

Variable	Definitions	Data Source
Information Events, Channels, and Queries (Firm-Day Level)		
<i>#Specific_Queries</i>	The number of firm-specific queries submitted on the platform.	Platform
<i>EA_Pos</i>	Indicator variable for a positive earnings announcement (i.e., an increase in net earnings relative to the previous accounting period) during the day.	CSMAR
<i>EA_Neg</i>	Indicator variable for a negative earnings announcement (i.e., a decrease in net earnings relative to the previous accounting period) during the day.	CSMAR
<i>MF_Pos</i>	Indicator variable for a positive management forecast (i.e., an increase in net earnings relative to the previous accounting period) during the day.	CSMAR
<i>MF_Neg</i>	Indicator variable for a negative management forecast (i.e., a decrease in net earnings relative to the previous accounting period) during the day.	CSMAR

Variable	Definitions	Data Source
<i>Event_Other</i>	Indicator variable for any other material events disclosed by the firm during the day.	CSMAR
<i>AnaReport</i>	Indicator variable for one or more analyst reports issued during the day.	CSMAR
<i>Media</i>	An indicator that the firm is mentioned in one or more news articles during the day.	CSMAR
<i>SVI</i>	Baidu search volume index for a given firm on the day.	Baidu
<i>Retail Platform</i>	The number of questions posted on public investor interactive platforms established by stock exchanges (Lee and Zhong, 2022).	CSMAR
<i>More_Topics</i>	An indicator variable that more than the sample median number of topics are discussed in that management forecasts.	CSMAR
<i>Longer</i>	An indicator variable that the management forecast is longer (as measured by its word count) than the sample median word count.	CSMAR
<i>PosFBRatio</i>	The proportion of answers that receive positive feedback (including thumbs-up, shares, or copies) out of all answers provided for a firm in a day.	Platform
<i>Avg_Sentiment</i>	The firm-day-level average sentiment (i.e., the proportion of positive sentences minus negative sentences) of all answers.	Platform
<i>Avg_Sentiment_PosFB</i>	The firm-day-level average sentiment of answers with positive feedback (including thumbs-up, shares, or copies).	Platform
<i>Avg_Sentiment_NonPosFB</i>	The firm-day-level average sentiment of answers without positive feedback (including thumbs-up, shares, or copies).	Platform
Trading Outcome Variables (Firm-Day Level)		
<i>AbnSpread</i>	The value-weighted average of the effective bid-ask spread scaled by the mid-quote, minus its average over trading days [-20,-1]. We multiply this variable by 100.	CSMAR
<i>Amihud Illiquidity</i>	The daily ratio of absolute stock return to its volume (Amihud, 2002). We multiply this variable by 100.	CSMAR

Variable	Definitions	Data Source
<i>AbnVol</i>	The daily stock turnover rate minus its average over trading days $[-20,-1]$. We multiply this variable by 100.	CSMAR
<i>VPIN</i>	The volume-synchronized probability of informed trading, calculated following (Easley et al., 2012). We multiply this variable by 100.	CSMAR
<i>AbnReturn</i>	The difference between the actual return and the benchmark return from a firm-specific time-series regression of daily stock returns on daily market returns over the $[-31,-1]$ trading-day window (Ke et al., 2003). We multiply this variable by 100.	CSMAR
<i>MV</i>	The firm's daily market value of equity.	CSMAR
Answer Attributes (Query/Answer Level)		
<i>Positive_FB</i>	Indicator variable that equals one if users give a thumbs-up, share, or copy a specific answer, and zero otherwise. We multiply this variable by 100.	Platform
<i>Thumbs_Down</i>	Indicator variable that equals one if users click thumbs-down for a specific answer, and zero otherwise. We multiply this variable by 100.	Platform
<i>#Future_Queries</i>	The number of future queries submitted by the user after the current query.	Platform
<i>Signal_Align</i>	Indicator variable that the trading signal embedded in the answer (identified using a commercial LLM) aligns with the firm's seven-day stock market performance.	Platform
<i>Forward_Looking</i>	The ratio of forward-looking sentences in the answer, as predicted by a fine-tuned Chinese BERT model.	Platform
<i>Subjectivity</i>	The ratio of subjective sentences in the answer, as predicted by a fine-tuned Chinese BERT model.	Platform

Variable	Definitions	Data Source
<i>Sentiment</i>	The net sentiment score of the answer, defined as the ratio of positive sentences minus the ratio of negative sentences, based on sentence-level predictions from a Chinese BERT model fine-tuned for sentiment classification, following Huang et al. (2023).	Platform
<i>Fog</i>	The Gunning-Fog index of the answer.	Platform
<i>Topic_Diversity</i>	The ratio of additional topics mentioned in the answer that were not asked for in the query, relative to the number of all topics not mentioned in the query.	Platform
<i>Specificity</i>	The number of specific words identified by the Stanford Stanza NER model in the answer, divided by the total number of words in the answer after stop-words removed, following Hope et al. (2016).	Platform
<i>#References</i>	The number of references in the answer.	Platform
<i>#Words</i>	The total number of words in the answer.	Platform
User Interaction and Query Patterns (User Level)		
<i>Specificity_First3_Queries (Answers)</i>	The specificity of the first three queries submitted by a given user (or of the answers to those queries), measured as the number of specific words scaled by total words.	Platform
<i>Specificity_Post3_Queries</i>	The specificity of queries from the fourth query onward submitted by a given user, measured as the number of specific words scaled by total words.	Platform
<i>Any_FM_First3_Queries (Answers)</i>	The average proportion of the first three queries (or the corresponding answers) that mention at least one financial metric.	Platform
<i>Any_FM_Post3_Queries</i>	The average proportion of queries that mention at least one financial metric from the fourth query onward.	Platform

Variable	Definitions	Data Source
<i>Specific_FM_First3_Queries (Answers)</i>	Three separate indicator variables equal to one if one of three specific financial metrics (ROE, Solvency, or Accounts Receivable) appears at least once in the first three queries (or in the answers to those queries), and zero otherwise.	Platform
<i>Specific_FM_Post3_Queries</i>	Three separate indicator variables equal to one if one of three specific financial metrics (ROE, Solvency, or Accounts Receivable) appears at least once from the fourth query onward, and zero otherwise.	Platform
<i>Attributes_Post3_Queries</i>	Linguistic attributes of queries submitted from the fourth query onward, including <i>Sentiment</i> , <i>Subjectivity</i> , <i>Forward_Looking</i> , and <i>#Words</i> , as well as the total number of subsequent queries.	Platform

APPENDIX B: QUERY TOPICS, TOP WORDS, AND ANSWER TOPICS

The table below presents the top 10 topics of user queries, along with their sample proportions, the top 15 words associated with each topic, and the corresponding topics mentioned in the elicited answers.²⁵ For the answer topics, the numbers in parentheses indicate the likelihood that a specific topic is mentioned in the answers. In total, we categorize queries and answers into 25 topics. The less frequent query topics not listed in the table below are: Regulation and Policy (4.93%), Corporate Internal Control (4.15%), Healthcare (3.99%), Banking (3.98%), Corporate Financing Decisions (3.42%), Corporate Restructuring and Financial Distress (3.19%), Supply Chain (2.98%), Macroeconomic Environment (2.88%), AI (2.31%), Board Governance and Executive Leadership (2.19%), Initial Public Offerings (IPO) and Delistings (2.04%), Media & Entertainment (1.38%), Human Capital, Management Team and Organization Culture (1.01%), CSR (0.94%), and Education Sector (0.26%).

²⁵ We used a TF-IDF-based approach to rank topic-specific keywords, addressing sentence overlaps across topics. Relative TF-IDF scores ensure distinctiveness by normalizing word importance within a topic against its overall sample importance.

Query Topic	Top 15 Words	Answer Topics
Business Outlook (26.32%)	Development Prospects, Prospects, Profit Forecast, Forecast, Tendency, Development Trend, Stock Price, Will Rise, Stock, Future, Investment, Industry, Value, Impact, Valuation	Business Outlook (78.54%), Product and Technology (69.58%), Competition (67.76%), Production and Operations (65.57%), Financial Performance (50.53%), Corporate Investment Decisions (42.80%), Customer and Brand (36.79%), Macroeconomic Environment (27.10%), Renewable Energy (23.51%), Supply Chain (19.70%)
Production and Operations (25.56%)	Operating Capacity, Core Business, Accounts Receivable, Operations, Inventory Turnover Ratio, Operating Revenue, Decline, Accounts Receivable Turnover Ratio, Total Asset Turnover Ratio, Inventory, Current Situation, Business, Capability, Research, Strategy	Production and Operations (78.11%), Product and Technology (60.74%), Business Outlook (59.34%), Competition (52.61%), Financial Performance (52.22%), Corporate Investment Decisions (37.91%), Customer and Brand (27.80%), Supply Chain (23.47%), Corporate Internal Control (18.67%), Renewable Energy (16.67%)
Financial Performance (23.26%)	Financial Statements, Financial Analysis, Net Profit, Financial Statement Analysis, Operating Revenue, Finance, Annual Report, ROE, Financial Condition, Balance Sheet, Growth Rate, Capability, Performance, Cash Flow Statement, Current Ratio	Financial Performance (85.30%), Production and Operations (52.65%), Business Outlook (46.48%), Competition (39.26%), Product and Technology (32.76%), Corporate Investment Decisions (29.85%), Macroeconomic Environment (17.63%), Corporate Internal Control (14.51%), Customer and Brand (13.91%), Shareholder Returns and Equity Changes (13.16%)
Product and Technology (22.26%)	R&D, Product, Battery, Core Business, Technology, Investment, Solid-state, Business, Field, Automobile, New Energy Vehicle, Competitive Advantage, Competitiveness, Core, Science and Technology	Product and Technology (90.11%), Business Outlook (70.11%), Production and Operations (66.91%), Competition (60.60%), Customer and Brand (38.79%), Corporate Investment Decisions (33.03%), Renewable Energy (29.60%), Financial Performance (26.39%), Supply Chain (20.73%), AI (20.43%)

Query Topic	Top 15 Words	Answer Topics
Stock Market Performance (19.60%)	Stock Price, Trend, Will Rise, Forecast, Tomorrow, Limit Up, Stock Trend, Buy, Decline, Market Value, Stock, Today, P/E ratio, Line Chart, Valuation	Business Outlook (62.16%), Financial Performance (54.78%), Competition (49.29%), Product and Technology (49.16%), Production and Operations (44.14%), Stock Market Performance (43.67%), Corporate Investment Decisions (28.70%), Macroeconomic Environment (27.17%), Customer and Brand (23.53%), Renewable Energy (19.75%)
Competition (18.04%)	Competitor, Position, Competition, Peer Industry, Weakness, Five Forces, Market Share (by value), Market Share (by volume), Industry, Market, Core, Advantage, Competitiveness, Impact, Competitive Advantage	Competition (80.34%), Product and Technology (76.40%), Business Outlook (74.56%), Production and Operations (63.36%), Customer and Brand (46.19%), Financial Performance (38.91%), Corporate Investment Decisions (34.04%), Renewable Energy (26.38%), Macroeconomic Environment (23.58%), Supply Chain (22.56%)
Corporate Investment Decisions (7.43%)	M&A, Acquisition, Suggestion, Worth, Strategic Analysis, Buy, Plan, Long-term Investment, Direction, Planning, Investment, Strategy, Value, Future, Tactics	Business Outlook (78.78%), Corporate Investment Decisions (77.12%), Production and Operations (66.84%), Product and Technology (65.70%), Competition (61.52%), Financial Performance (47.69%), Customer and Brand (28.10%), Macroeconomic Environment (26.68%), Regulation and Policy (22.92%), Shareholder Returns and Equity Changes (20.55%)
Renewable Energy (7.38%)	Battery, Renewable Energy, Solid-state, Energy, Photovoltaic, Energy Storage, DMI, Electric Vehicle, Lithium Battery, Sales, Power Battery, Technology, New Energy Vehicle, Automobile, Battery Swapping	Renewable Energy (86.40%), Product and Technology (83.09%), Business Outlook (75.14%), Competition (66.01%), Production and Operations (58.52%), Customer and Brand (39.61%), Financial Performance (33.49%), Corporate Investment Decisions (29.92%), Supply Chain (18.06%), Regulation and Policy (15.53%)

Query Topic	Top 15 Words	Answer Topics
Customer and Brand (6.45%)	Customer, Marketing, Channel, Sales, Brand, Price Reduction, Reliable, Sales Volume, Consumer, Customer Segment, Market, Battery Swapping, Strategy, Market Share (by value), Market Share (by volume)	Product and Technology (84.54%), Customer and Brand (80.97%), Competition (77.97%), Business Outlook (75.46%), Production and Operations (57.54%), Renewable Energy (31.99%), Financial Performance (31.64%), Supply Chain (24.48%), Corporate Investment Decisions (22.61%), Macroeconomic Environment (13.44%)
Shareholder Returns and Equity Changes (6.23%)	Shareholder, Equity Structure, Dividend Payout, Equity Incentive, Equity, Acquisition, Shares, Shareholding, Shareholding Reduction, Share Repurchase, Dividend Policy, Dividend, Holding, Dividend Yield, Shareholding Increase	Shareholder Returns and Equity Changes (84.50%), Business Outlook (32.76%), Financial Performance (30.95%), Corporate Investment Decisions (26.63%), Competition (20.42%), Production and Operations (19.05%), Stock Market Performance (18.14%), Product and Technology (17.45%), Board Governance and Executive Leadership (17.28%), Corporate Financing Decisions (15.51%)

APPENDIX C: TASKS, EXAMPLES, AND FREQUENCIES

This appendix presents the frequency distribution and examples of query subcategories related to the three information processing tasks for the subsample of 1,601,904 queries, corresponding to 91.7% of the total sample of 1,746,075 queries. The task categories are *not* mutually exclusive. The excluded observations in the data set are 144,155 queries (8.3%) with *Generative Tasks* such as user requests for GenAI reports or presentations and 52,176 queries (3.0%) with *Educational Tasks* such as questions about financial concepts or stock market fundamentals.

Information Processing Task Category and Examples	# of Queries
Information Awareness Tasks	677,665 (38.8%)
Stock Recommendation	156,683 (9.0%)
• <i>What are some trending stocks to watch in the A-share market?</i> • <i>Please recommend stocks related to NVIDIA Corporation.</i> • <i>Please recommend several stocks with relatively high dividend yields.</i> • <i>Can you recommend a stock for me to buy tomorrow?</i>	
News/Event Monitoring	309,322 (17.7%)
• <i>What is the latest news in the personal care and household products industry?</i> • <i>Did Hundsun Technologies Inc. release any major financial reports or announcements at that time?</i> • <i>What is the current state of R&D in the electric vehicle domain at SAIC Motor Corp., Ltd.?</i> • <i>What are the latest developments regarding China Merchants Bank Co., Ltd. in the past three months?</i>	
Information Acquisition Tasks	777,178 (44.5%)
Information Searching	570,735 (32.7%)
• <i>Please search for Kweichow Moutai Co., Ltd.'s 2023 annual report.</i> • <i>Can you find the current top 10 shareholders of Sinopec Corp. and their respective ownership percentages?</i> • <i>What was the return on assets (ROA) of Bank of China Limited and Postal Savings Bank of China Co., Ltd. in 2023?</i> • <i>Can you help me find a research report on the Chinese financial industry?</i>	
Cause Analysis	291,768 (16.7%)
• <i>Why has China Yangtze Power Co., Ltd.'s stock been performing so well and repeatedly hitting new highs?</i> • <i>Why is Kweichow Moutai Co., Ltd. so profitable?</i> • <i>Why did Midea Group Co., Ltd. choose to conduct its stock buyback through centralized bidding?</i> • <i>Why has BYD Co., Ltd. been successful?</i>	
Summarization	58,452 (3.3%)
• <i>What were the core messages and highlights of Hwa Create Co., Ltd.'s latest performance briefing?</i> • <i>What are the key takeaways from BYD Co., Ltd.'s 2023 annual report?</i> • <i>Please summarize the key risk management strategies highlighted in China Southern Airlines Company Limited's 2023 annual report and evaluate their effectiveness.</i> • <i>Please summarize the Q1 2024 financial report of Seres Group Co., Ltd.</i>	

Information Processing Task Category and Examples	# of Queries
Information Integration Tasks	1,354,301 (77.6%)
Comparison	235,552 (13.5%)
• <i>How do Midea Group Co., Ltd. and Gree Electric Appliances Inc. of Zhuhai differ in terms of their current strategic focus?</i> • <i>How does Dirui Industrial Co., Ltd. compare with its industry peers?</i> • <i>Which stock is better, Changchun Up Optotech Co., Ltd. or Fujian Casteck Crystals, Inc.?</i> • <i>How does the R&D expenditure of BYD Co., Ltd. compare with that of Tesla, Inc.?</i>	
Trend Prediction	238,112 (13.6%)
• <i>Will FiberHome Telecommunication Technologies Co., Ltd. expected to report higher earnings this year?</i> • <i>Can you predict the stock performance of Huaqin Chemical Co., Ltd. (A-share listed) for next week in 2024?</i> • <i>Which sectors may emerge as attractive investment opportunities in the second half of the year?</i> • <i>What are the future trends and regulatory support driving the development of the auto finance sector?</i>	
Impact Analysis	326,034 (18.7%)
• <i>What other factors could influence China Yangtze Power Co., Ltd.'s net profit margin?</i> • <i>How has digital transformation affected Haier Smart Home Co., Ltd.'s non-financial performance?</i> • <i>What impact will Zheshang Securities Co., Ltd.'s merger have on its business operations?</i>	
Overall Evaluation	277,820 (15.9%)
• <i>What is the overall assessment of CSSC Science & Technology Co., Ltd.'s 2023 financial health?</i> • <i>Is Juneyao Airlines Co., Ltd. a good company to invest in?</i> • <i>How did Jinneng Holding Shanxi Coal Industry Co.,Ltd. perform overall in 2023?</i>	
Financial Statement Analysis	255,969 (14.7%)
• <i>How strong was Zhejiang Supor Co., Ltd.'s debt-paying capacity in 2017?</i> • <i>How sound is Autel Intelligent Technology Corp., Ltd.'s capital structure based on its 2023 financial statements?</i> • <i>Analyze Changan Automobile Co., Ltd.'s financial reports over the past five years</i>	
Risk Assessment	89,889 (5.1%)
• <i>How does BYD Co., Ltd. prevent risks?</i> • <i>Is ST Gaosheng (Gaosheng Holding Co., Ltd., under special treatment) at risk of being delisted?</i> • <i>Current status of financial risk control at Jinke Property Group Co., Ltd.</i>	

Information Processing Task Category and Examples	# of Queries
Trading Strategy • Is it a good time to buy Duofuduo New Energy Technology Co., Ltd stock for short-term trading? • Is Hongfa Technology Co., Ltd. worth holding for the long term? • Can I buy CITIC Securities Co., Ltd. now?	52,343 (3.0%)

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